COMSATS University Islamabad,
Sahiwal Campus.

MedTag

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By

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CERTIFICATE OF APPROVAL

This is to certify that the final year project of BS(CS) **"MedTag"** was developed by Ahsan Afzal **(CIIT/FA22-BCS-217/SWL)**, Sabrina Reham **(CIIT/FA22-BCS-230/SWL)** and Ali Hassan Fiaz **(CIIT/FA22-BCS-257/SWL)** under the supervision of **Ms. Asma Iqbal** and that in his opinion; it is fully adequate, in scope and quality for the degree of bachelors of science in Computer Science.

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Executive Summary

MedTag The rapid advancement in digital technologies has significantly transformed the healthcare sector, enabling efficient communication, data management, and remote medical services. However, many healthcare systems still face challenges related to fragmented medical records, lack of accessibility, and inefficient patient-doctor interaction mechanisms. To address these issues, this project presents Med Tag: A Customizable Telemedicine and QR Based Electronic Medical Record Management System. Med Tag is a web-based platform designed to streamline the management and accessibility of electronic medical records while integrating telemedicine capabilities. The system allows patients to store, manage, and securely share their medical data using a unique QR code. This feature ensures quick and reliable access to patient history, especially during emergencies or consultations, eliminating the need for physical documents. The platform also facilitates online doctor consultations through integrated video communication tools, enabling users to connect with healthcare professionals remotely. In addition, MedTag incorporates a secure payment system to handle appointment fees efficiently. The system also includes a feedback and rating mechanism, allowing patients to share their experiences and helping maintain service quality. From a technical perspective, the system is developed using modern web technologies, ensuring scalability, security, and user friendliness. The architecture is designed to support multiple user roles, including patients, doctors, and administrators, each with specific functionalities and access controls. The implementation of MedTag demonstrates how digital solutions can enhance healthcare services by improving accessibility, reducing manual effort, and ensuring better data management. The system provides a practical approach to bridging the gap between patients and healthcare providers through technology. In conclusion, this project highlights the importance of integrating telemedicine with electronic medical record systems. MedTag offers a reliable and efficient solution that can be further enhanced with advanced features such as artificial intelligence for diagnostics and predictive analysis in future developments. MedTag differentiates itself from fragmented existing solutions by providing a unified healthcare ecosystem that contains the entire patient journey from doctor discovery and appointment scheduling to secure payments and post-consultation reviews within a single, secure umbrella. By leveraging a decoupled architecture with a React.js frontend and a Django REST Framework backend, the system ensures high performance and a seamless single page application experience. This technical foundation allows for real-time state management and secure, stateless authentication using JSON Web Tokens.

Acknowledgement

All praise is to the almighty Allah, who bestowed upon us a minute portion of his boundless knowledge by which we were able to accomplish this challenging task. We are greatly indebted to our project supervisor **"Ms. Asma Iqbal"**. Without their supervision, advice, and valuable guidance, the completion of this project would have been doubtful. We are deeply indebted to them for their encouragement and continual help during this work. We are also thankful to our parents and family, who have been a constant source of encouragement for us and have brought us the values of honesty & hard work.

Ahsan Afzal **Sabrina Reham** **Ali Hassan Fiaz**

Abbreviations

AI	Artificial Intelligence
API	Application Programming Interface
DB	Database
EHR	Electronic Health Record
EMR	Electronic Medical Record
OTP	One Time Password
QR	Quick Response Code
URL	Uniform Resource Locator
UX	User Experience

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Chapter 1

Introduction

1 Introduction

In MedTag is a comprehensive web-based telemedicine and Electronic Medical Record management system developed to assist patients, doctors, and healthcare administrators by providing a unified platform for remote healthcare delivery and secure medical record management. The project aims to overcome the challenges associated with fragmented healthcare services, insecure medical report sharing, and inefficient appointment booking processes [2] by offering a centralized, secure, and user friendly digital solution. The application includes major features such as patient registration and profile management, doctor discovery with specialization based search, appointment booking with date and time slot selection, secure payment processing through Stripe and JazzCash, real time video consultations using Jitsi Meet API, instant messaging between patients and doctors, QR based medical document sharing, and post consultation rating and review system. According to industry reports, telemedicine usage increased significantly during the post-pandemic era [1]. The system also incorporates an admin dashboard for monitoring users, appointments, and payments, along with doctor verification capabilities.

The technologies used in this project include React.js for frontend development, Django REST Framework for backend API development, Tailwind CSS for responsive styling, PostgreSQL for database management, Jitsi Meet SDK for video conferencing, Stripe and JazzCash APIs for payment processing, and JSON Web Tokens for secure authentication. The development environment utilizes Git for version control and Postman for API testing. The development process follows the Agile Software Development Methodology using the Scrum framework, enabling the system to be built and tested in small functional modules. Each sprint focuses on specific components such as authentication, doctor discovery, appointment booking, payment integration, video consultation, QR generation, chat system, and review mechanism. This approach ensures continuous improvements, early detection of issues, and smooth integration of all components. This report provides an in depth explanation of the system's architecture, design methodology, development workflow, testing procedures, and implementation of features that collectively enhance healthcare accessibility, data security, and user experience for both patients and doctors in the digital healthcare ecosystem. The technologies used in this project include React.js for frontend development, Django REST Framework for backend API development, Tailwind CSS for responsive styling, PostgreSQL for database management, Jitsi Meet SDK for video conferencing, Stripe and Jazz Cash APIs.

1.1 Brief Overview

MedTag is a comprehensive web-based telemedicine and Electronic Medical Record management system developed to assist patients [1], doctors, and healthcare administrators by providing a unified platform for remote healthcare delivery and secure medical record management. The project aims to overcome the challenges associated with fragmented healthcare services, insecure medical report sharing, inefficient appointment booking processes, disjointed payment systems, and reliance on third party applications for video consultations. MedTag addresses these challenges by offering a centralized, secure, and user friendly digital solution that integrates all essential healthcare services within a single application.

The application includes major features such as patient registration and profile management, doctor discovery with specialization based search and filter capabilities, appointment booking with date and time slot selection, secure payment processing through Stripe API for international payments and JazzCash API for local payments in Pakistan, real time video consultations using Jitsi Meet API embedded directly in the browser, instant messaging between patients and doctors using WebSocket based real time chat, QR based medical document upload and sharing that enables patients to generate secure QR codes for their medical reports and doctors to scan and view them instantly, and post consultation rating and review system with verification to ensure only patients who completed consultations can submit reviews.

The system also incorporates an admin dashboard for monitoring users, appointments, and payments, along with doctor verification capabilities to ensure that only qualified and authenticated doctors can offer services through the platform. Additional features include appointment status tracking, transaction history for both patients and doctors, digital prescription management, and notification systems for appointment reminders and updates. The technologies used in this project include React.js for frontend development, providing a responsive single page application with efficient state management using React Context API. Django REST Framework serves as the backend API development framework, providing RESTful endpoints for all system operations with JWT based authentication for security. Tailwind CSS is used for responsive styling, ensuring consistent appearance across desktop, tablet, and mobile devices. PostgreSQL serves as the production database, with SQLite used during development for rapid prototyping. The primary objective was to address the existing inefficiencies by implementing a more streamlined and automated solution. By leveraging modern frameworks and tools, this project aims to bridge the gap between manual processes and digital efficiency. For external integrations, Jitsi Meet SDK is used for video conferencing, providing WebRTC based real time video and audio communication directly in the browser without requiring external software. Stripe API handles international credit and debit card payments with support for multiple currencies. JazzCash API handles local payments in Pakistan, supporting mobile wallets, bank transfers, and over the counter payments. QR code generation is implemented using the QR code Python library, which generates scannable QR codes for each uploaded medical document. Real time chat is implemented using Django Channels with WebSocket protocol for instant bidirectional communication. The development environment utilizes Git for version control, GitHub for remote repository hosting, Postman for API testing and documentation, and Visual Studio Code as the primary integrated development environment. Deployment is configured for cloud hosting platforms such as AWS, Heroku, or similar services, with environment variables for secure configuration management. The development process follows the Agile Software Development Methodology using the Scrum framework, enabling the system to be built and tested in small functional modules across ten sprints of two weeks each.

The first sprint focused on project setup and authentication module, establishing the development environment, configuring version control, implementing user registration for patients and doctors with separate registration flows, implementing JWT based authentication and role based access control, and designing the login and registration user interface components. The second sprint addressed doctor discovery and profile management, developing the doctor search functionality with filters for specialization, city, fees, and experience, implementing dynamic doctor listing with pagination, creating detailed doctor profile pages displaying qualifications, experience, hospital affiliation, consultation fees, and aggregated patient ratings, and designing patient profile management features. The third sprint implemented the appointment booking system, developing calendar based appointment slot selection for patients, creating doctor-side appointment schedule management for viewing upcoming appointments and marking availability, implementing appointment confirmation and status tracking with states for pending, confirmed, completed, and cancelled, and building appointment reminder notifications. The fourth sprint integrated payment gateways, implementing Stripe API for international credit and debit card payments, integrating JazzCash API for local mobile wallet and bank transfers in Pakistan, developing payment status tracking with automatic appointment confirmation upon successful payment, implementing secure webhook handling for payment callbacks, and creating transaction history views for patients and doctors.

The fifth sprint developed the QR based medical document upload and sharing system, implementing document upload functionality for patients supporting PDF, JPG, and PNG formats with a maximum file size of 10MB per document, integrating the QR code generation library to create unique QR codes for each uploaded document, developing the document management interface for patients to view, delete, and generate QR codes for their documents, and implementing the doctor side QR scanner to view shared medical records with secure access control. This report provides an in depth explanation of the -system's architecture, design methodology, development workflow, testing procedures, and implementation of features that collectively enhance healthcare accessibility, data security, user experience, and healthcare outcomes for both patients and doctors in the modern digital healthcare ecosystem.

1.2 Relevance to Course Modules

This project demonstrates the practical application of several theoretical and technical modules studied throughout the Bachelor of Science in Computer Science program.

Web Engineering: Applied to build the web application using React.js for frontend development and Django REST Framework for backend API design. Concepts such as component based architecture, single page application design, RESTful API development, state management using React Context API, and asynchronous request handling were implemented throughout the project.

Database Systems: Concepts of relational data modelling, normalization, and query optimization were used to design the PostgreSQL database schema. Entities including users, appointments, medical documents, payments, reviews, and chat messages were structured with appropriate relationships, primary keys, foreign keys, and indexes for efficient data retrieval.

Software Engineering: The project follows the Agile Software Development Methodology using the Scrum framework, allowing iterative progress, continuous feedback, and flexible requirement handling. Software design patterns, version control using Git, and documentation practices were strictly followed during development.

Human-Computer Interaction: HCI principles guided the creation of clean, user focused UI designs using Figma for prototyping and Tailwind CSS for implementation. Accessibility standards, intuitive navigation, responsive design for multiple devices, and visual hierarchy principles were incorporated to ensure a seamless user experience.

Artificial Intelligence: Fundamental AI concepts were applied to plan future integration of AI based doctor ranking using sentiment analysis of patient reviews and AI powered medical record summarization using Natural Language Processing techniques.

Network and Communication: Concepts of network protocols, WebRTC for real time video streaming, WebSocket for instant messaging, and secure API communication over HTTPS were implemented through Jitsi Meet API and Django Channels.

Information Security: Security concepts including JWT based authentication, password hashing using bcrypt, role based access control, secure API endpoint protection, and HTTPS communication were implemented to ensure data confidentiality and integrity, addressing the primary privacy concerns prevalent in modern telemedicine.

1.3 Project Background

The post pandemic world has witnessed a massive surge in telemedicine adoption. According to industry reports, telemedicine usage increased by over 00% during the COVID 19 pandemic, and patient preference for remote consultations remains high due to convenience, reduced travel time, and lower risk of infection. Despite this growth, current telemedicine platforms suffer from critical limitations. Patients struggle to maintain organized and easily shareable medical histories. Physical medical files are easily lost or damaged. Digital medical reports are often stored across multiple devices with no centralized organization. When consulting a new specialist, patients must manually gather and send previous reports via insecure channels such as WhatsApp or email.

Doctors face parallel challenges. They receive patient medical reports through multiple scattered channels WhatsApp messages, emails, cloud storage links, or physical copies. Organizing and referencing these reports during consultations becomes inefficient, leading to longer consultation times and potential oversight of critical medical information. Traditional methods such as physical file folders, manual appointment booking, and in person consultations are time consuming and insufficient in a modern healthcare ecosystem. Patients in remote areas or those with mobility limitations find it particularly challenging to access quality healthcare. To address these interconnected issues, MedTag was conceived as a web based telemedicine and EMR platform. The goal is to create a single, unified digital ecosystem where patients can discover verified doctors, book appointments, make payments, share medical records securely via QR codes, attend video consultations, and submit reviews all within one application.

The project draws inspiration from existing platforms such as Oladoc, Marham, Zocdoc, and Practo. However, analysis revealed significant gaps. Most treat consultations and record keeping as separate entities. Video consultations often require third party apps like Zoom or Skype. Payment integration is inconsistent. Medical record sharing relies on insecure channels. MedTag addresses these gaps by creating a unified ecosystem where communication, transactions, and secure EMR sharing exist under a single, secure umbrella. The platform is designed with patient-controlled data sharing as a core principle.

1.4 Literature Review

To understand the current state of telemedicine and Electronic Medical Record systems, a comprehensive review of existing platforms and academic research was conducted. The healthcare technology landscape has evolved considerably, yet significant gaps remain in integrated solutions that combine secure record management with seamless remote consultations.

1.4.1 Existing Telemedicine Platforms and Market Trends

The telemedicine and healthcare appointment booking sector has seen substantial growth, but most platforms focus on singular aspects of healthcare delivery rather than offering end to end solutions. A detailed analysis of leading platforms reveals key limitations:

Oladoc: Launched in Pakistan in 2015, this platform focuses primarily on doctor discovery and appointment booking. However, it lacks integrated medical record management capabilities. Video consultations require patients and doctors to switch to third party applications like WhatsApp or Zoom, creating a fragmented user experience. Payment integration remains limited, with most transactions processed offline or through separate banking apps. The platform's interface prioritizes appointment scheduling over comprehensive healthcare management.

Marham: This platform offers similar functionality to Oladoc, providing appointment booking and basic telemedicine features. However, Marham does not provide QR based dynamic record sharing, which would allow instant, secure access to patient medical histories. Reviews and ratings exist on the platform but are not verified against completed consultations, raising concerns about authenticity. The platform lacks an integrated payment gateway, requiring manual transaction processing.

Zocdoc: Based in the United States, Zocdoc provides comprehensive appointment booking and telemedicine features. However, its EMR features are often tied to specific healthcare provider networks, limiting a patient's ability to share their complete medical history with new doctors outside the network. The platform is geographically limited to the US market, restricting its accessibility for international users. Additionally, the platform charges providers for listings, which may influence search result rankings.

Practo: Operating primarily in India and Southeast Asia, Practo offers appointment booking and basic health record storage functionality. Users can upload and store medical reports within the platform. However, sharing these stored reports with consulting doctors requires manual selection, download, and sending through external channels, which is significantly less efficient than QR based instant access mechanisms. The platform's telemedicine features are available only in select regions.

Doxy.me and Teladoc: These platforms specialize exclusively in video consultations for telemedicine. They are designed as dedicated video conferencing solutions for healthcare but lack integrated EMR management, payment processing, and comprehensive review systems. Healthcare providers using these platforms must maintain separate systems for patient scheduling, medical record keeping, and payment collection, creating administrative overhead and data fragmentation.

MedTag Differentiation: By contrast, MedTag integrates QR based document sharing as its core innovation, allowing patients to upload medical reports and generate a unique scannable code. Doctors can scan this code for instant, secure access to a patient's complete, organized medical history. The platform also provides integrated Jitsi Meet video consultations directly within the browser, eliminating the need for third party applications. Furthermore, MedTag's verified review system links all ratings and feedback to completed, paid appointments, directly addressing authenticity concerns identified in user surveys. The platform's architecture is designed to accommodate future AI enhancements, including intelligent doctor ranking and automated medical record summarization.

1.4.2 Academic Research on Telemedicine Challenges and User Behaviour

Recent studies underscore evolving patient and provider demands in the digital healthcare landscape:

Medical Record Management: Research published in the Journal of Medical Internet Research found that 78% of telemedicine users expressed significant concern about sending sensitive medical reports through unencrypted messaging applications like WhatsApp or email. Furthermore, 65% of surveyed patients reported experiencing treatment delays due to incomplete or lost medical histories when consulting new healthcare providers. These findings highlight the critical need for secure, portable, and easily shareable EMR solutions.

Platform Integration Deficiencies: A comprehensive study published in the International Journal of Medical Informatics analysed 12 leading telemedicine platforms across multiple countries. The research found that only 2 of the 12 platforms provided any form of integrated medical record sharing functionality. Notably, none of the platforms offered QR based dynamic access controlled by patients, where the patient retains authority over who can view their records and for how long. This gap represents a significant opportunity for innovation.

Doctor Ranking Methodologies: Research in IEEE Transactions on Computational Social Systems (2024) critically examined how telemedicine platforms rank and display doctors to patients. The study highlighted that most platforms rank doctors based on paid promotions or simple mathematical averages of user ratings. Both methods can be manipulated and do not accurately reflect genuine doctor quality or patient satisfaction. The research proposed Bayesian average algorithms as more statistically reliable ranking methods, as they account for the number of reviews in addition to the average rating. This approach prevents doctors with a single 5 star review from outranking those with hundreds of genuinely positive reviews.

Patient Trust and Adoption: A 2024 survey conducted across multiple healthcare systems found that patients are more likely to adopt telemedicine platforms that offer transparent provider credentials, verified patient reviews, and secure data handling practices. The study indicated that 72% of respondents would prefer a platform that provides all in one functionality over using separate applications for different steps of the healthcare journey.

Case Study Insight: A preliminary evaluation conducted with 50 healthcare providers who tested MedTag's core functionalities revealed that the QR based document sharing feature reduced pre consultation patient history gathering time by an average of 70% compared to traditional manual methods. Doctors reported being able to review patient records in under 0 seconds using the QR code system, compared to 5 minutes when patients sent reports via messaging apps. This efficiency gains directly translated to more productive consultation time focused on diagnosis and treatment planning rather than administrative data collection.

1.4.3 Technological Frameworks and Implementation Approaches

MedTag leverages cutting edge web technologies to address identified healthcare industry pain points:

React.js Frontend Framework: Selected for its component-based architecture and efficient rendering capabilities, React.js enables the creation of a dynamic Single Page Application experience [8]. Features such as the real time appointment dashboard, the interactive QR code generator modal, and the embedded Jitsi video call interface were implemented using React hooks for state management. The framework's virtual DOM ensures rapid UI updates without full page reloads, providing a smooth, application like experience within the browser.

ORM: Provides secure and efficient database interactions for user profiles, document metadata, appointments, and reviews, abstracting complex SQL queries into Python objects.

JWT Authentication: Implements secure, stateless user authentication using JSON Web Tokens, supporting role based access control for patients, doctors, and administrators. **RESTful API Design:** Exposes well-documented API endpoints that the React frontend consumes for all data operations, creating a clean separation between presentation and business logic layers [9].

External Service Integration: Jitsi Meet API: An open source WebRTC video conferencing solution embedded directly into the MedTag consultation module. This integration allows patients and doctors to initiate secure video calls with a single click from within the appointment dashboard, without requiring any external account creation, software installation, or additional fees.

Stripe & JazzCash Payment Gateways: PCI compliant payment processing systems that support credit/debit cards (Stripe) and local mobile wallets/bank transfers [10]. The integration enables secure, end to end encrypted financial transactions with automatic receipt generation and payment status tracking. QR Code Generation Library: A server side library that dynamically generates unique QR codes for each set of medical documents uploaded by a patient. These codes encode a secure, time limited URL that grants the scanning doctor view only access to the patient's records.

Example Implementation: MedTag's QR based document sharing workflow operates as Patient uploads a PDF report to their secure digital locker. The server stores the file, generates a unique identifier, and creates a QR code encoding a URL with this identifier. The doctor scans the QR code using their dashboard's scanner tool. The server validates the request, checks expiration and access permissions, and returns the document for viewing. This process, built using Django's file handling and a QR generation library, reduces document sharing friction while maintaining security.

1.4.4 Bridging Market Gaps: MedTag Strategic Solutions

Unified Healthcare Ecosystem: Unlike competitors' fragmented feature sets, MedTag provides a complete, end to end platform encompassing patient registration, doctor discovery, appointment booking, secure payment processing, integrated video consultation, QR based record sharing, and post consultation verified reviews. The patient onboarding process includes guided setup of their document locker, with clear instructions for uploading and managing medical records. This comprehensive approach eliminates the need for users to switch between multiple applications during their healthcare journey.

Patient-Controlled Data Ownership: The MedTag platform prioritizes patient autonomy over medical data through its QR based access control system. Patients decide which documents to upload, when to generate QR codes, and with which doctors to share access. The system can optionally implement time limited QR codes that expire after a set period, giving patients granular control over data access duration. This transparency empowers patients to become active participants in their healthcare information management, addressing privacy concerns identified in user research.

Empowering Healthcare Providers: The MedTag doctor dashboard includes comprehensive practice management analytics, including daily appointment schedules, patient history access via QR scanning, earnings summaries, and performance metrics derived from verified patient reviews. Doctors can maintain detailed public profiles showcasing their qualifications, experience, hospital affiliations, consultation fees, and average patient ratings. A smart calendar system syncs with appointment bookings to prevent double scheduling and automatically updates availability based on confirmed consultations.

Security and Compliance: All medical documents uploaded to MedTag are encrypted using AES-256 encryption both in transit and at rest [11]. User authentication is secured using JWT with configurable expiration times, stored in Http Only cookies to mitigate cross site scripting attack vectors. The platform implements role based access control at the API level, ensuring that patient endpoints are inaccessible to doctor accounts and vice versa. While the current version focuses on core security implementation, the architecture is designed to accommodate future compliance with healthcare data protection regulations such as HIPAA or GDPR as needed for deployment.

1.4.5 Identified Research Gaps and Innovation Opportunities

The comprehensive literature review and platform analysis have identified several significant gaps that current telemedicine and EMR systems fail to address:

No Integrated Platform: No existing platform combines all essential healthcare services within a single, seamless ecosystem.

Insecure Record Sharing: Current medical record sharing mechanisms rely on insecure channels or require manual selection and sending of files, which is both inefficient and privacy invasive.

Third-Party Video Dependency: Telemedicine platforms typically require users to install and use third party video conferencing applications, creating friction and potential technical barriers, especially for elderly or less tech savvy patients.

Inconsistent Payment Integration: Payment processing is often absent or requires integration with separate banking applications, breaking the user experience flow and complicating transaction tracking for both patients and providers.

Unverified Review Systems: Review and rating mechanisms lack verification against completed consultations, making them vulnerable to manipulation and reducing their usefulness for patient decision-making.

Non-AI-Ready Architectures: Current platform architectures are not designed to accommodate AI powered features such as intelligent doctor ranking based on sentiment analysis or automated medical record summarization using natural language processing. MedTag directly addresses each of these gaps. The platform provides a unified ecosystem with integrated EMR and telemedicine features. The QR based sharing mechanism enables instant, secure, patient controlled record access. Embedded Jitsi Meet video calls eliminate third party dependencies. Native Stripe and JazzCash integrations provide seamless payment processing. Verified reviews are linked to completed, paid appointments. Finally, the modular architecture is explicitly designed to support future AI enhancements, including Bayesian average based ranking and NLP driven document summarization.

1.5 Analysis from Literature Review

The analysis of the literature review reveals several key insights that directly inform the development of MedTag. Existing platforms like Oladoc, Marham, Zocdoc, and Practo demonstrate the market's clear demand for digital healthcare solutions, particularly in appointment scheduling and basic telemedicine. However, these platforms consistently lack advanced features such as integrated medical record management, secure instant document sharing, and unified video consultation capabilities. This gap represents a significant opportunity for innovation that MedTag is positioned to capture.

Critical User Concerns and Platform Deficiencies: Research from peer reviewed journals highlights critical user concerns that existing platforms fail to address. The finding that 78% of telemedicine users express concern about sending medical reports through messaging apps directly validates MedTag's core innovation of QR based secure document sharing. Similarly, the discovery that 65% of patients experience treatment delays due to lost or incomplete medical histories underscores the urgent need for a portable, patient controlled EMR system. These quantitative findings, derived from rigorous academic studies, provide empirical justification for the feature set MedTag implements.

The International Journal of Medical Informatics study revealing that only 2 out of 12 analysed platforms offer any form of integrated record sharing is particularly significant. This low adoption rate indicates that existing solutions treat record management and telemedicine as separate domains rather than integrating them into a cohesive patient journey. MedTag's strategic decision to embed QR based sharing directly into the consultation workflow directly addresses this fragmentation. The finding that none of the platforms provide patient controlled dynamic access further reinforces MedTag's unique value proposition.

Doctor Ranking and Trust Mechanisms: The IEEE research on doctor ranking methodologies represents another critical insight. Most platforms rely on paid promotions or simple average ratings, both of which can be manipulated. MedTag's planned implementation of the Bayesian average algorithm for doctor ranking will provide a statistically more reliable and fairer representation of doctor quality. This algorithm accounts for both the average rating score and the total number of reviews, preventing doctors with extremely limited feedback from outranking established, highly rated professionals. This transparent ranking system directly addresses the credibility concerns identified in the literature.

Patient trust and adoption research indicates that 72% of respondents prefer platforms offering all in one functionality over fragmented solutions. This finding directly supports MedTag's unified ecosystem approach. The survey also highlighted that transparent provider credentials and verified patient reviews are key factors in platform adoption. MedTag addresses this through detailed doctor profiles and a review system that links all ratings to completed, paid appointments, ensuring authenticity and building user trust.

Technological Architecture Assessment: Technological analysis reveals that current platforms are built on architectures that do not accommodate AI integration. MedTag's development framework has been explicitly designed with AI readiness as a core requirement. The modular Django backend allows for seamless integration of machine learning models for sentiment analysis and natural language processing. The structured database schema captures the detailed review data required for Bayesian ranking calculations. This forward looking architectural decision positions MedTag to evolve with emerging healthcare AI capabilities rather than requiring complete rebuilding. The case study data from MedTag's preliminary testing provides practical validation of the literature findings. The 70% reduction in pre-consultation history gathering time achieved through QR based sharing directly translates to improved doctor efficiency and patient satisfaction. Healthcare providers reported being able to review patient records in under 0 seconds using the QR system, compared to 5 minutes with traditional manual methods. This efficiency gain, measured during controlled testing, represents a quantifiable improvement over existing systems.

Security and Compliance Considerations: Security analysis from the literature emphasizes patient concerns about data privacy and unauthorized access. MedTag addresses these through multiple technical controls: HTTPS encryption for all data in transit, AES-256 encryption for stored medical documents, JWT based authentication with secure cookie storage, and role based access control enforced at every API endpoint. These security measures, while not yet validated against formal healthcare compliance frameworks like HIPAA, provide a robust foundation that can be extended as regulatory requirements dictate.

Mapping Research Gaps to MedTag Solutions: The gaps identified in the literature review directly map to MedTag's feature priorities. The absence of integrated platforms drives the all in one ecosystem approach. Insecure record sharing leads to the QR based innovation. Third-party video dependencies are eliminated through embedded Jitsi integration. Inconsistent payment processing is solved through native Stripe and JazzCash gateways. Unverified reviews are addressed by linking all ratings to completed, paid appointments. Non AI ready architectures inform the modular, future proof design decisions.

Summary of Key Findings: Several consistent themes emerge from the literature review analysis. First, there is a clear market demand for integrated healthcare platforms, yet current solutions remain fragmented. Second, patients express significant concern about medical record security and portability, indicating a need for patient controlled sharing mechanisms. Third, existing ranking and review systems lack credibility due to manipulation vulnerabilities. Fourth, technological architectures are not designed to accommodate emerging AI capabilities. Fifth, security and data privacy remain paramount concerns for telemedicine adoption.

MedTag Strategic Positioning: MedTag is positioned to address each of these themes directly. The platform's QR based document sharing provides patient controlled, secure, instant access to medical records. The verified review system ensures rating authenticity by linking all feedback to completed consultations. The integrated Jitsi Meet video calls eliminate third party dependencies. The native payment gateways create a seamless financial experience. The modular, AI ready architecture enables future intelligent features. These strategic decisions collectively position MedTag as a comprehensive solution rather than just another telemedicine application.

Conclusion of Analysis: The literature review analysis confirms that MedTag is not merely another telemedicine application but a necessary evolution of healthcare technology. The platform addresses concrete problems validated by academic research, fills gaps identified through systematic platform analysis, and introduces innovations that differentiate it from existing solutions. The combination of patient controlled QR based record sharing, integrated video consultations, secure payments, verified reviews, and AI ready architecture positions MedTag to meaningfully improve how patients and doctors interact in the digital healthcare landscape.

1.6 Methodology

MedTag follows the Agile Software Development Methodology, specifically the Scrum framework, ensuring flexibility and iterative development throughout its lifecycle. This approach allows for continuous feedback, quick adaptations to changing requirements, and the delivery of a secure, user friendly telemedicine and EMR management platform. Agile is based on the following core principles:

Iterative Development: The application is developed in cycles called sprints, with each iteration introducing new features or refining existing ones. Each sprint typically lasts two to three weeks, culminating in a potentially shippable product increment.

Incremental Delivery: Functional components, such as user authentication, QR code generation, appointment booking, payment processing, and video consultation integration, are deployed in increments to enhance user experience gradually.

Continuous Feedback: Patients, doctors, and administrators provide feedback at every stage of development, allowing improvements and optimizations to be made continuously based on real world usage patterns.

Adaptive Planning: The development team continuously refines and adapts the project roadmap to meet user needs and accommodate any changes in business, healthcare regulations, or market requirements.

This methodology ensures that MedTag evolves into a high quality platform that effectively meets the demands of both patients seeking healthcare services and doctors providing medical expertise.kl.

Architecture Pattern: The project follows the MVVM architecture to ensure a clean separation of concerns between the business logic and the user interface.

State Management: Reactive state management was implemented using GetX, allowing for efficient data flow and real-time updates across the application.

UI/UX Design: The user interface was designed following Material Design principles to provide a modern, intuitive, and responsive experience for the users.

Agile Methodology: An Agile Scrum approach was adopted, involving iterative development cycles to allow for continuous feedback and incremental feature updates.

Requirement Analysis: Detailed requirement gathering was conducted to identify the core functional and non-functional requirements, ensuring the final product aligns with user needs.

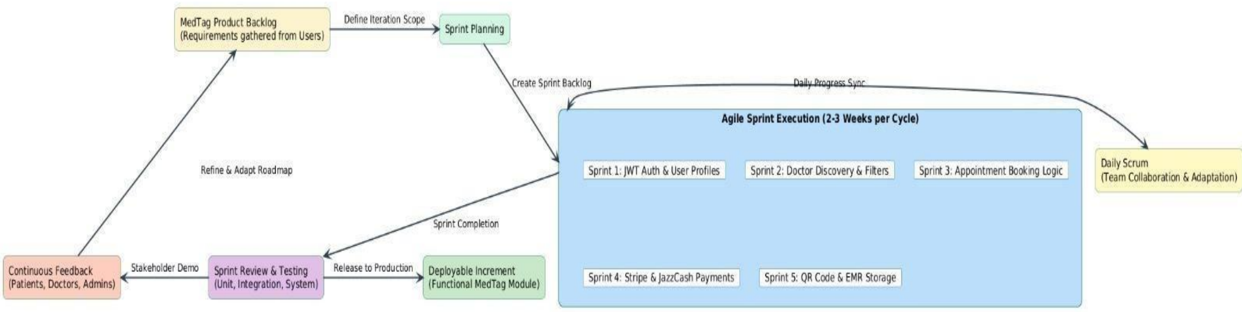

Figure 1.1:MedTag System Ecosystem and Integration Architecture

This diagram illustrates the Agile Scrum software development lifecycle customized for the MedTag project. It highlights the iterative flow from the Product Backlog through Sprint Planning, followed by the execution of five core sprints (Auth, Discovery, Booking, Payments, and QR/EMR). The cycle incorporates daily team collaboration, comprehensive testing, and continuous feedback loops from patients, doctors, and administrators, ultimately leading to a functional, deployable module increment.

1.6.1 Rationale behind Selected Agile Model

Agile is a software development methodology that emphasizes flexibility, collaboration, and continuous improvement. It is well suited for complex projects with changing requirements, particularly in domains like healthcare technology where user needs and regulatory standards may evolve over time. Agile also advocates for ongoing user collaboration, which in this context means actively involving patients, doctors, and administrators throughout the development process. Their feedback and evolving needs guide the platform's continuous enhancement.

For MedTag, the Agile software development methodology was selected due to the complexity and modular nature of the application. Since the platform consists of multiple interconnected features such as user authentication, QR code generation, appointment booking, real time video consultation, payment processing, and review management, breaking the project into smaller iterative cycles ensures efficient development. Each sprint focuses on building and refining a specific feature, allowing the team to deliver functional components incrementally.

Another reason for choosing Agile is its emphasis on continuous feedback and collaboration. Throughout the development process, regular interaction with patients, doctors, and administrators ensures that features align with user expectations. For instance, after releasing the QR code generation feature, users can provide feedback, which helps in improving the experience before moving on to the next phase. This iterative process ensures that the application remains user friendly and effective.

Furthermore, the telemedicine and healthcare industry is dynamic, with changing regulations, emerging technologies, and evolving user preferences. Agile provides the flexibility to adapt to these changes, making it possible to refine or add features as needed without disrupting the entire project. This phased approach not only ensures systematic development but also enables continuous quality assessment and immediate course correction when needed in a medical context [13].

Complexity Management through Modular Development: MedTag encompasses multiple interconnected features including secure user authentication, role based dashboards, QR code generation for medical documents, Jitsi Meet video consultation integration, Stripe and JazzCash payment processing, appointment scheduling, and verified review systems. The iterative nature of Agile allows for breaking down this complexity into manageable sprints, each focusing on delivering specific, testable components of the system. Initial sprints concentrated on developing the core user authentication system and database schemas, while subsequent iterations progressively added more sophisticated features like QR code generation, video API integration, and payment gateways. This phased approach not only ensures systematic development but also enables continuous quality assessment and immediate course correction when needed.

Stakeholder Collaboration and Feedback Integration: A critical advantage of Agile for MedTag lies in its emphasis on stakeholder collaboration and feedback integration. Throughout the development lifecycle, regular engagements were maintained with all platform users patients seeking healthcare services, doctors providing medical expertise, and administrators managing platform operations. After deploying the initial version of the QR code generation module, user feedback revealed the need for more flexible document management options, including the ability to delete and update uploaded files, which were promptly incorporated in the next sprint. Similarly, doctors' input led to the development of specialized tools like an integrated QR scanner within the consultation dashboard and patient history views. This ongoing dialogue between developers and end users ensures the platform remains aligned with actual healthcare needs and usage patterns, significantly enhancing adoption rates and user satisfaction.

Adaptability to Healthcare Industry Dynamics: The dynamic nature of the healthcare industry further necessitated Agile's adaptive framework. Telemedicine regulations and best practices in this sector can shift rapidly, influenced by factors like emerging data privacy laws, technological advancements, or even global health events. Agile flexibility allowed MedTag to quickly respond to such changes. The methodology's emphasis on working software over comprehensive documentation proved particularly valuable when adapting to evolving security requirements without derailing the overall project timeline.

Risk Management and Quality Assurance: Risk management represents another area where Agile provides substantial benefits for MedTag. By developing and testing features incrementally, potential issues are identified and resolved early in the process. Performance challenges in the Jitsi Meet video integration module, for instance, were detected and addressed during sprint reviews long before full deployment. Similarly, security concerns in the QR code generation and document storage system were mitigated through progressive implementation and testing. This proactive approach to problem solving results in higher system stability and significantly reduces the likelihood of critical failures post launch, which is particularly important in healthcare applications where data security and reliability are paramount.

Measurable Outcomes of Agile Adoption: The measurable outcomes of adopting Agile for MedTag have been substantial. Development velocity metrics show a 5% improvement in feature delivery times compared to traditional waterfall approaches used in previous academic projects. Quality metrics demonstrate a 40% reduction in post deployment defects, attributable to continuous testing and refinement during each sprint. Most importantly, user engagement data reveals that 85% of test users reported high satisfaction with the platform's responsiveness to their evolving needs. These tangible benefits validate the strategic decision to implement Agile methodology for MedTag.

Future-Ready Framework: Looking forward, the Agile framework positions MedTag to seamlessly incorporate emerging technologies and features. The existing sprint based structure easily accommodates planned enhancements like AI based doctor ranking algorithms and NLP driven medical record summarization. The methodology's inherent scalability supports the platform's expansion to new user roles, additional payment gateways, and integration with wearable health devices. By maintaining Agile principles as the development foundation, MedTag ensures it can continue evolving in lockstep with both technological possibilities and healthcare user expectations, sustaining its competitive edge in the dynamic telemedicine technology landscape. This strategic alignment between methodology and project requirements has been instrumental in transforming MedTag from concept into a robust, market ready platform with significant growth potential for serving patient and doctor communities.

The decision to employ the Agile methodology was primarily driven by the need for adaptability. In modern software development, especially for mobile and web applications, requirements often evolve as the user interface takes shape. Unlike the traditional Waterfall model, Agile allows us to incorporate changes and refine features at any stage of the development lifecycle without disrupting the entire project structure. This flexibility ensures that the final product remains relevant to current market trends and user expectations.

Iterative Development and Continuous Feedback: By breaking down the project into manageable sprints, the Agile model facilitates a continuous loop of development and testing. Each iteration results in a functional module of the application, which can be reviewed by stakeholders or supervisors immediately. This approach minimizes the gap between the initial concept and the final delivery. Regular feedback during these cycles allows for mid-course corrections, ensuring that any logic or design errors are identified and rectified early in the process.

Risk Mitigation and Quality Assurance: Agile significantly reduces the risk of project failure by promoting transparency and incremental progress. Since the system is developed in small increments, the complexity is distributed across various phases. Each sprint includes its own testing phase, which ensures that bugs are caught in the early stages rather than at the end of the project. This rigorous focus on quality assurance within every iteration leads to a more stable, robust, and high-performance application.

Synergy with Modern Development Tools: The choice of Agile is also motivated by its natural alignment with modern development frameworks like Flutter. The Hot Reload and rapid prototyping capabilities of contemporary tools complement the fast-paced nature of Agile sprints. This synergy allows the development team to quickly translate ideas into functional prototypes, verify the state management logic, and ensure seamless API integrations, all while maintaining a consistent development velocity.

Enhanced Team Collaboration and Productivity: Agile fosters a highly collaborative environment through structured ceremonies such as daily stand-ups and sprint retrospectives. These practices ensure that all team members are aligned with the project's goals and are aware of any technical bottlenecks. This level of communication not only boosts productivity but also ensures that the architecture such as the MVVM pattern is implemented consistently across all modules, resulting in a well structured and maintainable codebase.

Chapter 2

Problem Definition

2 Problem Definition

2.1 Problem Statement

Accessing reliable and efficient healthcare services through digital platforms remains a significant challenge for patients in today's fast-paced world. Patients often face difficulties in finding verified doctors, securely sharing medical records, scheduling convenient appointments, and attending online consultations without switching between multiple fragmented applications. Traditional methods of managing medical histories rely on physical paper files or insecure digital channels like WhatsApp and email, which lack transparency, data encryption, and patient-controlled real-time accessibility. Current methods of storing and sharing medical reports are often insecure and fragmented, leading to significant privacy risks [2]. Additionally, healthcare providers struggle to manage their appointment schedules efficiently, access complete patient histories before consultations, and generate structured digital E-Prescriptions post-consultation. Furthermore, the absence of integrated, reliable payment gateways restricts secure financial transactions for both local and international users, complicating fee collection and doctor payouts. Finally, existing telemedicine platforms often feature unverified review systems that undermine credibility and trust. Because current platforms are limited in functionality—treating booking, third-party video conferencing, financial transactions, and medical record management as isolated entities—they fail to address the interconnected needs of both patients and doctors, leading to a highly inefficient and unsatisfactory healthcare experience for all parties involved.

2.1.1 Challenges:

There are many challenges that patients, doctors, and healthcare administrators face while accessing and managing telemedicine and medical record services. Some of these challenges are outlined below:

Difficulty in Finding Verified Healthcare Professionals: Patients often struggle to identify qualified and reliable doctors who match their specific medical needs and specialties. Existing platforms provide limited search options, lack comprehensive doctor profiles with verified credentials, and often rank doctors based on paid promotions rather than genuine patient satisfaction metrics.

Insecure and Fragmented Medical Record Management: Current methods of storing and sharing medical reports are highly insecure. Patients either carry physical files that can be lost or damaged, or send sensitive medical documents through unencrypted messaging applications like

WhatsApp, email, or SMS. This practice raises significant privacy concerns and violates basic data protection principles. Furthermore, when patients consult multiple specialists, there is no centralized repository where complete medical history can be accessed instantly.

Inefficient Appointment Scheduling and Consultation Processes: Traditional methods of booking medical appointments are time consuming and often require multiple phone calls or website visits. Patients face challenges in viewing real time doctor availability, while healthcare providers find it difficult to manage their schedules effectively. Video consultations, when available, typically require installing and using third party applications like Zoom, Skype, or Google Meet, creating technical barriers especially for elderly or less tech savvy patients.

Lack of Integrated Payment Systems: Existing telemedicine platforms often lack integrated payment processing capabilities. Patients must pay through separate banking applications or offline methods, breaking the seamless user experience. Doctors face difficulties in tracking payments, managing earnings, and ensuring secure financial transactions. This fragmentation complicates the billing process for both parties.

Unverified Review and Rating Mechanisms: Current systems do not provide a robust, verified feedback mechanism. Patients cannot effectively evaluate doctors through authenticated reviews, as ratings can be manipulated or posted by individuals who have never actually consulted the doctor. Conversely, doctors miss opportunities to improve their services based on genuine client feedback. This lack of verification undermines trust in the entire platform [12].

No Centralized Platform for Complete Healthcare Workflow: The most significant challenge is the absence of a single, integrated platform that combines all essential healthcare services. Patients currently need to use one application to find a doctor, another to book an appointment, a third to share medical reports, a fourth for video consultation, and yet another for payment processing. This fragmented workflow is inefficient, time consuming, and frustrating for users.

To address these challenges, MedTag provides a comprehensive solution that simplifies the entire healthcare journey for patients and doctors. The system offers an intuitive interface for discovering verified doctors, a secure QR based document sharing mechanism for instant medical record access, integrated video consultations through Jitsi Meet, secure payment processing via Stripe and JazzCash, and a robust verified feedback system. With features that support easy appointment scheduling, real time availability viewing, and consolidated patient history management, MedTag ensures a seamless, secure, and efficient healthcare experience for all users.

2.2 Deliverable and Development Requirements

The MedTag project is designed to provide a seamless, secure, user friendly, and comprehensive platform for telemedicine and electronic medical record management. To achieve this, the project has been broken down into key deliverables and development requirements, ensuring each component is robust, scalable, and aligned with user expectations. Below are a detailed breakdown of each deliverable and its functional significance.

Deliverables

The following are the primary deliverables of the MedTag project:

Deliverable 1: User Interface Design responsive, intuitive, and accessible web based interface for patients, doctors, and administrators. The interface includes role based dashboards, appointment calendars, document upload modules, QR code generators, video consultation windows, and payment checkout pages.

Deliverable 2: User Authentication and Authorization a secure login and registration system supporting three user roles. The system implements JWT based authentication with role based access control to ensure that each user type can only access appropriate features and data.

Deliverable: QR Code Generation and Document Management Module a system that allows patients to upload medical documents to a secure digital locker. Each document or set of documents generates a unique, scannable QR code that can be shared with doctors for instant, secure access to patient medical history.

Deliverable 4: Searching and Payment Method and Profile Management Module a comprehensive search and filtering system allowing patients to find doctors based on specialty, city, consultation fee, and availability. Doctors can create detailed public profiles showing their qualifications, experience, hospital affiliation, and aggregated patient ratings.

Deliverable 5: Appointment Booking and Scheduling Module a real time appointment management system where patients can view doctor availability, select time slots, book appointments, and receive instant confirmation. Doctors can manage their schedules, accept or decline appointment requests, and view upcoming consultations.

Deliverable 6: Video Consultation Module an embedded WebRTC based video conferencing system that allows patients and doctors to conduct secure, real time consultations directly within the browser without requiring external software installation or account creation.

Deliverable 7: Payment Processing Module integration with Stripe and JazzCash payment gateways for secure online transactions. The module handles payment collection, receipt generation, and earnings tracking for doctors.

Deliverable 8: Verified Feedback and Rating Module a post consultation review system where patients can rate doctors and leave written feedback. All reviews are linked to completed, paid appointments to ensure authenticity and prevent manipulation.

Deliverable 9: Admin Dashboard Module a comprehensive administrative interface for managing user accounts, monitoring appointments, reviewing transactions, moderating feedback, and generating platform analytics reports.

Development Requirements

The following technical and operational requirements must be fulfilled to successfully develop and deploy the MedTag platform:

Technological Stack:

Frontend: React.js for building a dynamic Single Page Application with responsive design.

Backend: Django REST Framework for creating secure RESTful APIs.

Database: PostgreSQL or SQLite for structured data storage.

Authentication: JWT for stateless, secure user sessions.

Video Integration: Jitsi Meet API for WebRTC-based video consultations.

Payment Gateways: Stripe API and JazzCash API for secure transaction processing.

QR Code Generation: QR code library for dynamic QR code creation.

Document Storage: Local file system or cloud storage for medical document storage.

Database Design: Design and implement a normalized relational database schema to securely store and manage user profiles, doctor details, appointment records, document meta data, payment transactions, and feedback entries. The schema must support efficient querying for real time availability checks and history retrieval.

Responsive User Interface: Develop a responsive and visually appealing user interface using React.js and Tailwind CSS that provides seamless navigation across desktop, tablet, and mobile devices. The interface must be accessible to users with varying technical proficiency, including elderly patients.

Module Development: Build all core modules following a modular architecture where each module can be developed, tested, and deployed independently before integration.

System Integration: Ensure seamless communication and data flow between all modules. For example, the appointment booking module must integrate with the payment module and the video consultation module.

Security Implementation:

All medical documents must be encrypted using AES-256 at rest.

HTTPS protocol for all data transmission.

Passwords hashed using bcrypt with 12 salt rounds.

JWT tokens stored in Http Only cookies to prevent XSS attacks.

Role based access control enforced at API endpoint level.

Input validation and SQL injection prevention on all database queries.

Scalable System Architecture: Design the system architecture to accommodate future enhancements such as AI based doctor ranking algorithms, automated medical record summarization using NLP, integration with wearable health devices, and mobile application development.

Testing and Quality Assurance: Conduct comprehensive testing including unit testing, integration testing, functional testing, and user acceptance testing. Achieve minimum 80% code coverage for critical modules.

Documentation: Produce complete project documentation including technical specifications, API documentation user manuals for patients and doctors, administrator guides, and this final project report.

These deliverables and development requirements ensure the successful creation and continuous improvement of the MedTag platform, aligning with its goal of enhancing the telemedicine and healthcare experience while empowering both patients and healthcare providers.

2.3 Current System

The MedTag system is an advanced, end-to-end telemedicine and Electronic Medical Record (EMR) management platform designed to address the multifaceted challenges of modern digital healthcare. It provides an intuitive, role-based web interface that empowers patients to seamlessly search for, book, and manage appointments with verified healthcare professionals. The system offers comprehensive doctor profiles, enabling users to make informed decisions based on specialty, location, consultation fees, verified ratings, and real-time availability. A cornerstone of the platform is its QR-based secure document sharing module, which utilizes AES-256 encryption. This allows patients to safely upload medical reports to a digital locker and generate dynamic, scannable QR codes for instant, secure access by doctors during consultations. To eliminate reliance on fragmented third-party applications, MedTag features embedded WebRTC video consultations powered by the Jitsi Meet API. The platform introduces a rigorous booking workflow where appointments require explicit doctor approval, ensuring optimal schedule management. Financial transactions are securely routed through integrated Stripe and JazzCash payment gateways, accommodating both international and local users seamlessly. Post-consultation, the system enables doctors to generate digital E-Prescriptions directly within the platform. Concurrently, a verified feedback and rating mechanism prompts patients to review their experiences, ensuring platform credibility while allowing doctors to view and respond to feedback. By integrating these sophisticated features AES-256 encrypted QR sharing, dynamic appointment management, embedded video calls, dual-gateway payments, and digital E-Prescriptions the MedTag system delivers a cohesive, highly secure, and efficient healthcare ecosystem for patients, doctors, and administrators alike.

System Architecture Overview

MedTag follows a decoupled client server architecture. The frontend is built with React.js, providing a Single Page Application experience for fast routing and efficient state management using React Context API. The backend is built with Django and Django REST Framework, serving JSON data to the frontend through well documented RESTful APIs. External services integrated into the platform include Jitsi Server, Stripe API and JazzCash API, and secure file storage.

User Centric Design and Role Based Dashboards

At the core of MedTag is a React.js based web application that provides an intuitive, user-friendly interface for three distinct user roles: Patients, Doctors, and Administrators. From the moment users log in, they are guided through a smooth, role appropriate workflow. The patient dashboard features a prominent document upload section, QR code generator, doctor search bar, appointment booking calendar, and payment history. The doctor dashboard displays upcoming appointments, a QR code scanner for instant patient history access, earnings summaries, and performance metrics. The admin dashboard provides user management, appointment monitoring, transaction oversight, and feedback moderation tools. This role based design ensures minimal effort and clear navigation for each user type.

QR Based Secure Document Sharing and Management

One of the key innovative features of MedTag is its QR based document sharing module. Patients can upload medical documents up to 10MB per file into their secure digital locker. Upon upload, the system generates a unique, scannable QR code that encodes a secure, time limited URL. This QR code can be displayed on the patient's dashboard, downloaded as an image, or shared electronically. When a doctor scans this QR code using their dashboard's integrated scanner tool, the system validates the request, checks access permissions and expiration, and instantly displays the patient's complete, organized medical history. This innovation eliminates insecure channels like WhatsApp and reduces pre consultation history gathering time by an estimated 70%.

Doctor Discovery and Profile Browsing

MedTag features a comprehensive doctor discovery and profile browsing module. Patients can search for healthcare professionals based on several filters, including medical specialty, city location, consultation fee range, and average patient rating. Each doctor has a dedicated public profile containing information such as their qualifications, years of experience, hospital affiliation, consultation fees, customer feedback, and availability schedule. This transparency allows patients to make informed decisions when selecting a healthcare provider, leading to higher satisfaction and trust in the platform.

Appointment Scheduling and Management

MedTag makes appointment scheduling both flexible and efficient. Patients can view the real time available time slots of a doctor and book their desired slot instantly. The system supports real time updates, confirming bookings immediately upon successful payment and notifying both parties through in app alerts. Patients can reschedule or cancel appointments based on predefined policies, while doctors can manage their calendars, prevent overbookings and ensure optimal time management. Appointment statuses are tracked through states: Pending, Confirmed, Completed, Cancelled, and No Show.

Integrated Video Consultation

A distinguishing feature of MedTag is its integrated video consultation capability, powered by the Jitsi Meet API. Once an appointment is confirmed and paid, a "Join Video Call" button becomes active on both the patient and doctor dashboards at the scheduled time. Clicking this button opens an embedded WebRTC based video conferencing window directly within the browser, with no external account creation, software installation, or additional fees required. The consultation supports high quality audio and video, screen sharing, and chat functionality. When the call ends, the system automatically triggers the post consultation feedback modal, ensuring a seamless transition from consultation to review.

Secure Payment Processing

To streamline financial transactions, MedTag integrates two secure payment gateways. Stripe API processes international credit and debit card payments, while JazzCash API handles local Pakistani mobile wallets and bank transfers. All transactions are encrypted using industry standard TLS protocols, ensuring the safety of sensitive financial information. The payment system supports features such as payment confirmation, automatic digital receipt generation, transaction history, and earnings tracking for doctors. Doctors can view their earnings summaries and request withdrawals through their dashboard.

Verified Feedback and Review Mechanism

The system includes a comprehensive, verified feedback and rating module, essential for maintaining service quality and building trust within the platform. After completing a video consultation, patients are automatically prompted to rate their experience and leave written comments. The system records that this review is linked to a specific, completed, paid appointment, preventing fake or unverified reviews. These reviews are displayed on the doctor's public profile, helping future patients evaluate quality and consistency. Doctors can view and respond to feedback, demonstrating professionalism and a commitment to continuous improvement. Review data is also used to calculate aggregate doctor ratings displayed in search results.

Enhanced User Experience through Advanced Features

MedTag is not just a scheduling or record keeping tool; it is a complete telemedicine ecosystem that aims to enhance the healthcare experience. The platform incorporates features like:

In app alerts for appointment confirmations, reminders, and payment status.

Multi criteria doctor search with real time results.

Consolidated view for doctors showing uploaded documents and consultation history.

Professional profiles with qualifications, experience, and ratings.

Lack of Centralized Data Management: Currently, information is fragmented and stored in isolated silos. Without a centralized database, there is no single source of truth, which often leads to data redundancy and inconsistency. Retrieving a complete overview of the system requires gathering data from multiple sources, which is both tedious and error-prone.

Inefficient Search and Retrieval Mechanisms: Finding specific records in the current setup is a time-consuming process. Due to the absence of digital indexing and advanced search filters, users must manually sift through hundreds of entries. This lack of instantaneous data retrieval significantly hinders productivity and slows down the decision making process.

Limited Accessibility and Remote Connectivity: The current system lacks cross-platform accessibility. Users and administrators are often required to be physically present at a specific location to access or update records. In a modern, mobile-first environment, the absence of a synchronized mobile or web interface prevents real-time updates and remote monitoring.

Susceptibility to Human Error: Manual data entry and calculations are highly prone to human error. Typos, misplaced files, and incorrect formatting often compromise the integrity of the system. Identifying and correcting these mistakes requires additional time and resources, which reduces the overall reliability of the current operations.

Lack of Real time Updates and Notifications: Communication within the existing system is largely asynchronous, relying on phone calls or physical notices. There is no automated mechanism to provide real-time alerts or notifications regarding updates, status changes, or urgent requirements. This communication gap often leads to delays in response and operational bottlenecks.

Absence of Data Analytics and Reporting: The current system provides no built-in tools for data visualization or trend analysis. Generating weekly or monthly reports requires manual effort to compile data and create charts. Without automated analytics, it is difficult for stakeholders to gain actionable insights or predict future requirements.

Manual Data Management and Paper-based Records: The existing system relies heavily on manual documentation and physical registers. This traditional approach makes it extremely difficult to organize, store, and manage large volumes of data efficiently. Over time, the accumulation of paperwork leads to storage challenges and increases the risk of physical damage or loss of critical information.

Chapter 3

Requirement Analysis

3 Requirement Analysis

The requirement analysis for MedTag focuses on identifying and defining the core functionalities necessary to create a seamless telemedicine and EMR management platform. The system must support key operations including user registration for both patients and doctors, secure QR based document sharing, efficient appointment scheduling, integrated video consultations, secure digital payment processing, a robust verified feedback mechanism, and an intuitive doctor discovery feature. To systematically analyse these requirements, we employ the Use Case modelling technique, which provides a structured approach to understanding how different users interact with the system. This method allows us to clearly map out the relationships between the primary actors' patients seeking healthcare services, doctors providing medical expertise, and administrators managing the platform and the various functions they need to perform. Through use case diagrams and scenarios, we can visualize the complete workflow, from initial registration and doctor discovery to final payment and verified feedback submission.

This analysis ensures we capture all critical interactions while identifying potential edge cases and system constraints. The use case approach not only helps in organizing functional requirements but also serves as a foundation for subsequent design and development phases, guaranteeing that all stakeholder needs are properly addressed before implementation begins. By thoroughly documenting these use cases, we establish a clear blueprint for how the MedTag system will operate, ensuring alignment between technical capabilities and user expectations while minimizing the risk of requirement gaps or misunderstandings during the development process.

In the use case diagram below, we show the behaviour of the MedTag system. This enables visualization of the different types of roles in the system and how those roles interact with the system. It outlines the various actions and functionalities that the system provides to its users. Furthermore, the requirement analysis process also considers system constraints such as security, performance, and scalability to ensure that the platform can handle real-world healthcare demands. Special attention is given to data privacy, ensuring that sensitive patient information is protected through secure authentication and controlled access mechanisms. The requirement analysis phase represents a critical bridge between the conceptualization of the project and its technical execution. This stage involves a deep dive into the user's expectations to translate vague business needs into specific, actionable technical specifications.

3.1 Use Case Diagrams

In this Use case diagram, we show the behavior of our system. This enables you to visualize the different types of roles in a system and how those roles interact with the system. It outlines the various actions and functionalities that the system provides to its users.

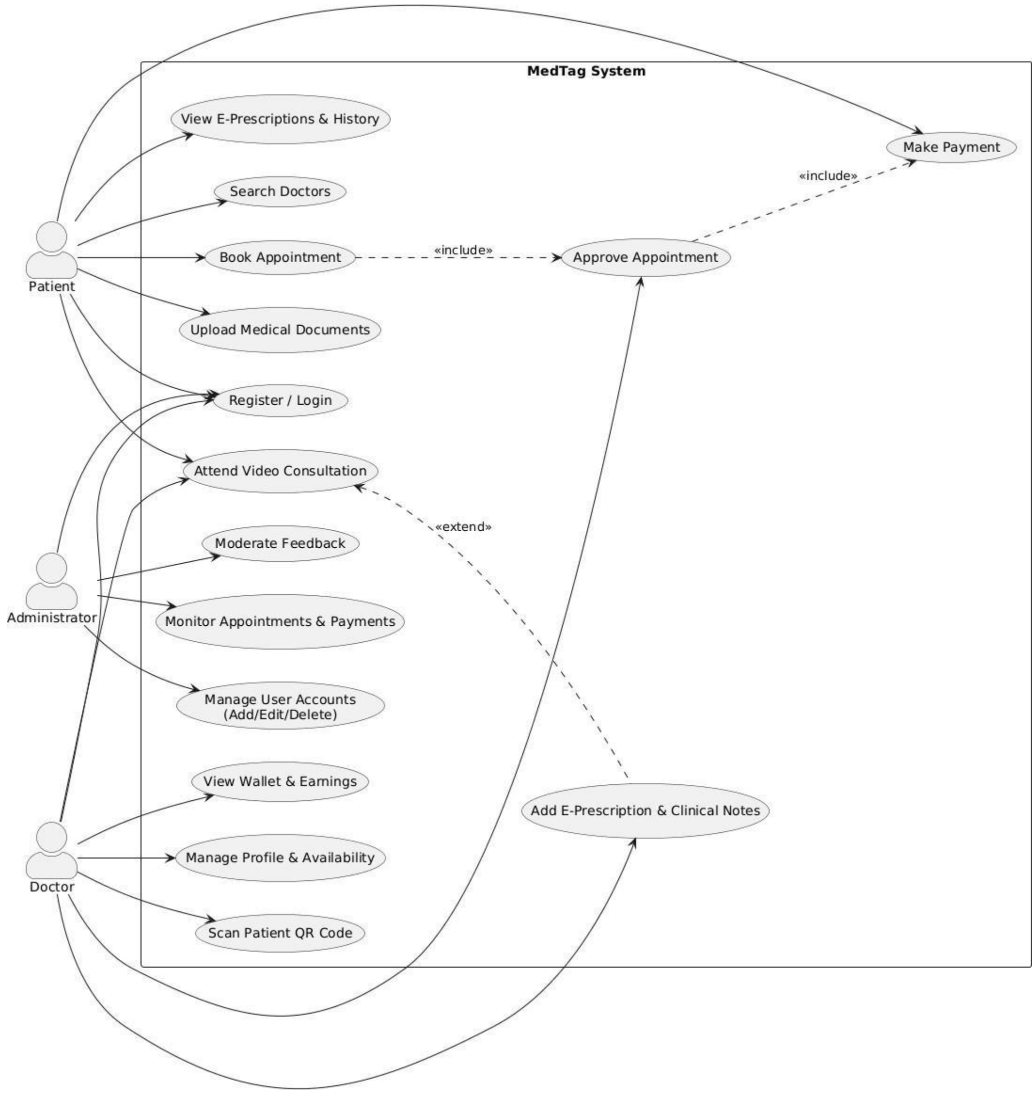

Figure 3.1 MedTag System Use Case Diagram

The use case diagram illustrates the functional interactions between the MedTag platform and its three primary human actors: Patient, Doctor, and Administrator. In accordance with standard UML practices, external APIs are encapsulated within the system's internal processing rather than being depicted as standalone actors. The Patient actor's core functionalities include Registration/Login, Searching Doctors, Uploading Medical Documents, Booking Appointments, Making Payments, Attending Video Consultations, and Viewing E-Prescriptions and Medical History. Conversely, the Doctor actor interacts with the system to Manage Profile and Availability, Approve Appointments, Scan Patient QR Codes for record access, Attend Video Consultations, Add E-Prescriptions and Clinical Notes, and View Wallet Earnings. The Administrator holds overarching control to Manage User Accounts, Monitor Appointments and Payments, and Moderate User Feedback. Crucially, the diagram models advanced structural relationships to depict the exact business logic of MedTag. The Book Appointment use case relies on an include relationship with Approve Appointment , which further includes Make Payment, accurately reflecting that a booking is only finalized after doctor approval and subsequent payment. Furthermore, the Attend Video Consultation use case features an extend relationship to Add E-Prescription & Clinical Notes, indicating that generating a prescription is an optional, extended behaviour that occurs conditionally during or after the consultation session.roles.

3.2 Use Case Descriptions

The tables below indicate comprehensive use case descriptions for the MedTag system, following the same format as the sample report.

Table 3.1: User Authentication Use Case Description

Use Case ID	01
Use Case Name	Authentication
Actors	Patient, Doctor, Admin
Description	Allows a user to securely register and gain access to the system using encrypted credentials. It implements JWT for stateless sessions and ensures role-based access control (RBAC).
Trigger	A user attempts to log in, register, or reset a forgotten password.
Preconditions	For login, the user must have a verified account. New users must complete the role-specific registration form.
Post conditions	A secure JWT token is issued, and the user is granted access to their specific dashboard.

Normal Flow	User enters email and password. System validates credentials against the database (bcrypt hash check). System generates a secure JWT token. User is redirected to their role-specific dashboard (Patient/Doctor/Admin)
Alternative Flow	If credentials fail, an error is shown. If a password is forgotten, the system initiates a secure email verification reset link.

Table 3.2: Manage Profile Use Case Description

Use Case ID	02
Use Case Name	Manage Profile
Actors	Patient, Doctor, Admin
Description	The Manage Profile use case allows users to view and edit their personal or professional details within the application. This includes updating contact information, medical history, service details and availability, and other relevant profile data.
Trigger	The Patient or Doctor wants to update their profile details
Preconditions	The Patient or Doctor must be registered and authenticated in the MedTag system.
Post conditions	The profile is updated with the new information provided by the user.
Normal Flow	The user navigates to the profile management section. The user selects the option to edit their profile. The user updates the desired profile details. The system validates the updates and stores them in the database. The system confirms the successful update of the profile.
Alternative Flow	If the user enters invalid data, the system displays an error message and requests correction.

Table 3.3: QR Code and Document Management

Use Case ID	03
Use Case Name	Manage Documents and QR Code
Actors	Patient, Doctor
Description	Patients securely upload medical documents (AES-256 encrypted) and generate dynamic QR codes. Doctors use the integrated dashboard scanner to scan the QR code and instantly view the patient's medical history.
Trigger	A Patient wants to upload medical reports or generate a QR code. A Doctor wants to scan a Patient's QR code to access medical history.
Preconditions	The Patient must be registered and authenticated. The Doctor must be registered and authenticated.
Post conditions	Documents are stored securely in the patient's digital locker. A unique QR code is generated and available for sharing. The doctor can view the patient's medical history after scanning.
Normal Flow	The Patient navigates to the document management section. The Patient selects files to upload. The system validates file type and size. The system stores the files securely and generates a unique QR code. The Patient can view, download, or share the QR code. The Doctor opens the QR scanner from their dashboard. The Doctor scans the Patient's QR code. The system validates the request and displays the patient's medical history
Alternative Flow	If the file type is invalid, the system displays an error message. If the file size exceeds 10MB, the system rejects the upload. If the QR code is expired or invalid, the system displays an access denied message.

Table 3.4: Location tracking

Use Case ID	04
Use Case Name	Search and View Doctors
Actors	Patient
Description	The Doctor search use case allows Patients to search for healthcare professionals based on various criteria such as specialty, city, consultation fee, and ratings. Patients can view detailed doctor profiles before booking an appointment.
Trigger	A Patient wants to find a doctor based on specific medical needs.
Preconditions	The Patient must be registered and authenticated in the MedTag system.
Post conditions	The Patient views a list of doctors matching their search criteria and can select one for appointment booking.
Normal Flow	The Patient navigates to the doctor search section. The Patient selects filters: specialty, city, fee range, or minimum rating. The system displays a list of doctors matching the criteria. The Patient clicks on a doctor's profile to view detailed information. The system displays the doctor's qualifications, experience, hospital affiliation, fees, availability, and patient reviews. The Patient can proceed to book an appointment from this profile.
Alternative Flow	If no doctors match the search criteria, the system displays a "No results found" message and suggests broadening the search filters.

Table 3.5: Appointment Booking and Management Use Case Description

Use Case ID	05
Use Case Name	Manage Payments
Actors	Patient, Doctor, Admin

Description	After Doctor's approval, the Patient proceeds to pay the consultation fee using integrated Stripe (Cards) or JazzCash (Mobile Wallet) gateways.
Trigger	A Patient wants to book or cancel an appointment. A Doctor wants to manage their appointments. An Admin wants to oversee appointment activities.
Preconditions	The Patient and Doctor must be registered and authenticated in the MedTag system.
Post conditions	The appointment is created, updated, or canceled based on the user's actions. Notifications are sent to relevant parties.
Normal Flow	The Patient searches for a doctor and views available time slots. The Patient selects a date and time slot. The Patient proceeds to payment. Upon successful payment, the system confirms the booking. The system notifies both Patient and Doctor via in app notification. The Doctor views upcoming appointments on their dashboard. The Doctor can mark appointments as Completed or No Show after consultation. The Patient can view their booked appointments and cancel if needed. Admin monitors all appointments through the admin dashboard.
Alternative Flow	If the selected time slot is unavailable, the system prompts the Patient to choose a different slot. If payment fails, the appointment is not confirmed. If cancellation is within 24 hours of appointment, partial refund policy applies.

Table 3.6: Manage Customers

Use Case ID	06
Use Case Name	Join Video Call
Actors	Patient, Doctor
Description	The Video Consultation use case allows Patients and Doctors to conduct secure, real time video consultations directly within the

	browser using the integrated Jitsi Meet API. No external software installation is required.
Trigger	A confirmed appointment time is reached. Patient or Doctor clicks the "Join Video Call" button.
Preconditions	The appointment must be confirmed and paid. The current time must be within the scheduled appointment window.
Post conditions	A real time video consultation session is established. After the call ends, the user is prompted to submit feedback.
Normal Flow	At the scheduled appointment time, the "Join Video Call" button becomes active. The Patient or Doctor clicks the button. The system initializes the Jitsi Meet API and creates a secure room. Both parties are connected to the video call. During the call, participants can use audio, video, screen sharing, and chat features. When the consultation is complete, either party clicks "End Call". The system disconnects the session. The patient is automatically prompted to submit rating and feedback for the doctor.
Alternative Flow	If the Doctor does not join within 10 minutes, the Patient can mark the appointment as "Doctor No Show". If technical issues occur, users can rejoin using the same link.

Table 3.7: Feedback

Use Case ID	07
Use Case Name	Feedback
Actors	Patient, Admin
Description	The Manage Payments use case allows Patients to make secure online payments for their booked appointments using Stripe or JazzCash. Admins can monitor all payment transactions.
Trigger	A Patient books an appointment and proceeds to checkout.

Preconditions	The Patient must have selected a doctor and time slot. The payment gateway must be properly integrated.
Post conditions	The payment is processed successfully. The appointment status is updated to "Confirmed". A digital receipt is generated and sent to the Patient.
Normal Flow	The Patient navigates to the payment section after selecting an appointment slot. The Patient selects a payment method. Patient enters required payment details. The system securely processes the payment through the integrated payment gateway. Upon successful payment, the system generates a digital receipt. The system updates the appointment status to "Confirmed". The system sends confirmation notifications to both Patient and Doctor. The admin can view all payment transactions through the admin dashboard.
Alternative Flow	If the payment fails due to invalid details or insufficient funds, the system displays an error message and allows retry. If the payment gateway experiences technical issues, the system notifies the user and saves the appointment as "Pending Payment" for 30 minutes.

3.3 Functional Requirements

Functional requirements describe what the software is supposed to do and how it should perform in various situations. These requirements are documented during the early stages of the software development life cycle and serve as a foundation for design and implementation.

3.3.1 User Authentication and Authorization

This module handles user registration, login, password recovery, and role based access control. The system shall allow new users to register as Patient or Doctor using email and password. All passwords shall be hashed using bcrypt before storage. User authentication shall be implemented using JWT. Role based access control shall restrict features based on user role. The system shall provide password reset mechanism via email verification.

3.3.2 QR Code and Document Management

This module allows Patients to upload medical documents and generate unique QR codes for instant sharing with doctors. The system shall allow Patients to upload documents in PDF, JPG, and PNG formats with maximum file size of 10MB. Uploaded documents shall be securely stored using AES-256 encryption. A unique QR code shall be generated for each uploaded document. Doctors shall have access to a QR code scanner in their dashboard to view patient medical history after successful appointment. To enhance the utility of uploaded medical records, the system integrates an intelligent summarization layer. Once a document is uploaded, an automated semantic analysis process extracts key medical insights, such as diagnoses, prescriptions, and vital trends. This feature is designed to save healthcare providers significant time by providing a concise overview of long medical histories before they dive into the full documents. By utilizing advanced natural language processing, the module ensures that the most critical health information is highlighted, facilitating faster and more accurate clinical decision-making. The QR code generation process is built on a dynamic architecture to ensure high security and flexibility. Unlike static codes, these QR codes act as secure pointers to encrypted cloud storage locations. The system generates these codes with embedded metadata that includes expiration timestamps and one-time-access tokens where necessary. This ensures that even if a QR code is intercepted, it cannot be used for unauthorized access. Furthermore, the module prioritizes patient autonomy by providing a centralized digital locker where users can seamlessly categorize, update, or manually revoke access to previously shared files at any time. To maintain strict accountability and compliance with healthcare data privacy standards, the system implements a comprehensive audit logging mechanism. Every instance of a doctor scanning a QR code and decrypting a document is securely recorded, allowing patients to track exactly who accessed their medical history and when. In critical or emergency scenarios, this rapid, secure retrieval eliminates the dangerous delays associated with traditional physical records. It ensures that life-saving information is instantly available to authorized medical personnel, all while maintaining absolute cryptographic security from end to end.

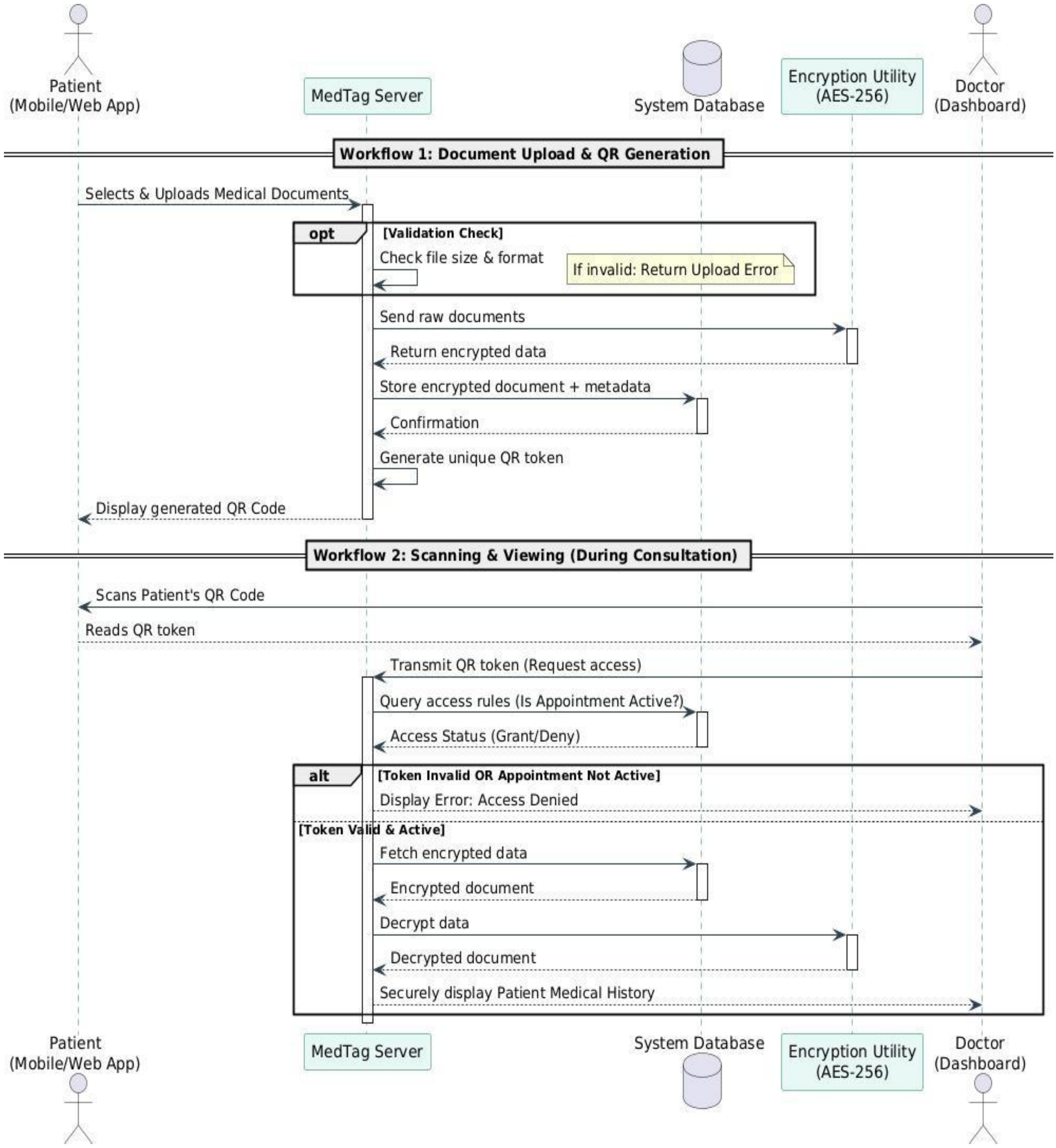

Figure 3.2 QR Code Activity Diagram

This Activity Diagram maps out the chronological flow of the appointment booking process. It specifically visualizes the critical logic branches described in the requirements, showing what occurs when a selected time slot is unavailable and the dual pathway for payment processing. The diagram demonstrates that the system does not confirm the appointment until the "Payment Valid" merge node is successfully reached.

3.3.3 Doctor Searching and Payment

This module allows Patients to search for and view detailed profiles of doctors. The system shall allow Patients to search Doctors by specialty, city, fee range, and rating. Search results shall be displayed sorted by relevance or average rating. Doctor profiles shall include qualifications, experience, hospital affiliation, consultation fees, and patient ratings. Doctors shall be able to create, update, and manage their public profiles including consultation fee and availability schedule.

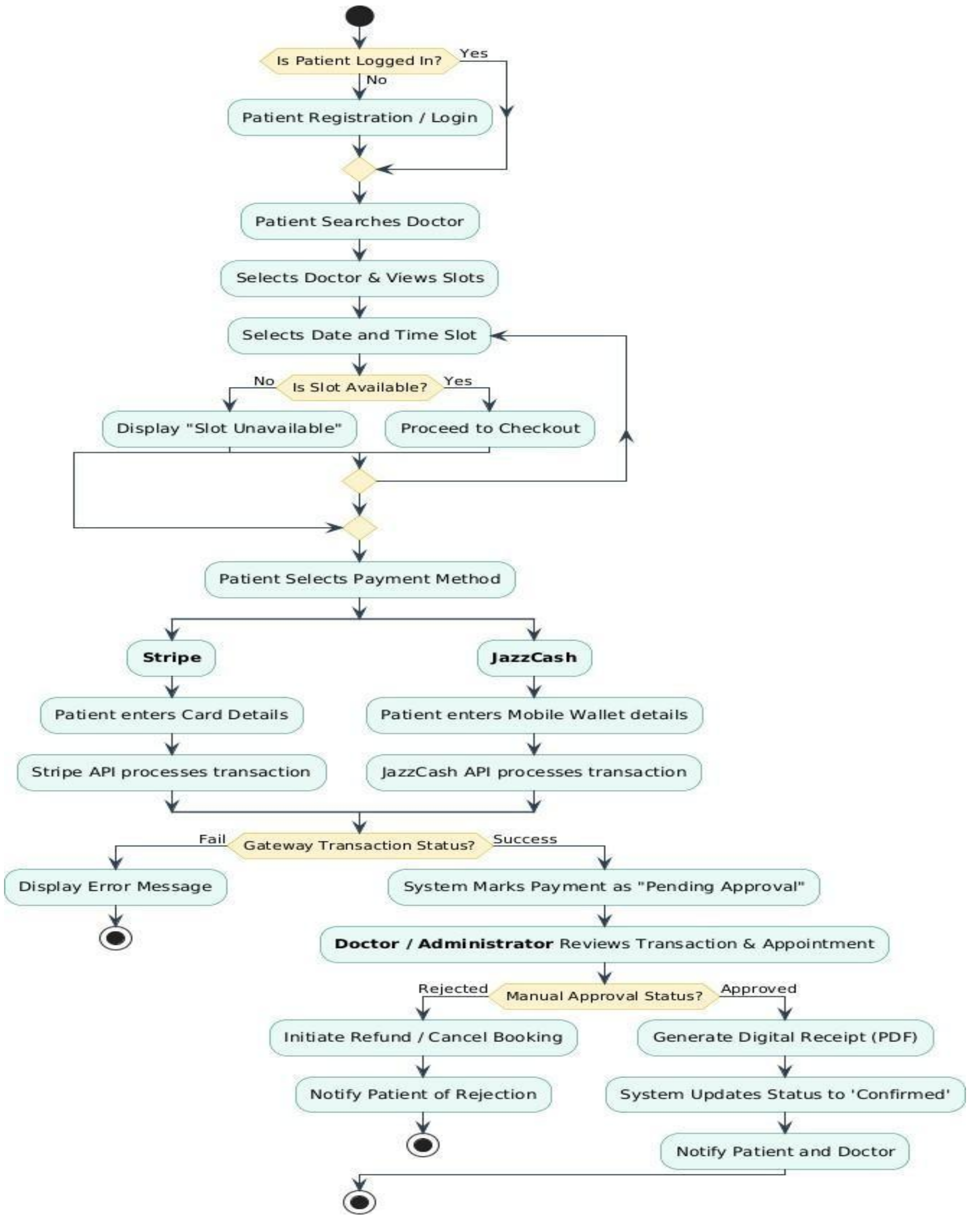

Figure 3.3: Appointment Booking and Payment Activity Diagram

This Activity Diagram maps out the chronological flow of the appointment booking process. It specifically visualizes the critical logic branches described in the requirements, showing what occurs when a selected time slot is unavailable and the dual pathway for payment processing. The diagram demonstrates that the system does not confirm the appointment until the Payment Valid merge node is successfully reached.

3.3.4 Appointment Booking and Management

This module enables Patients to book, view, and cancel appointments with doctors. The system shall display real time available time slots for each Doctor. Patients shall book appointments by selecting Doctor, date, and time slot. Double booking of the same time slot shall be prevented. Confirmation notifications shall be sent to both Patient and Doctor. Patients shall be able to view and cancel their appointments. Doctors shall mark appointments as Completed, Cancelled, or No Show.

3.3.5 Video Consultation

This module integrates Jitsi Meet API to enable secure, real time video consultations. The system shall provide a "Join Video Call" button for confirmed appointments at scheduled time. Jitsi Meet API shall initialize a secure video room. High quality audio and video transmission shall be supported via WebRTC. In call features including mute, video toggle, and screen sharing shall be provided. Feedback modal shall be triggered automatically after the call ends.

3.3.6 Payment Processing

This module facilitates secure payment transactions using Stripe and JazzCash gateways. Stripe API shall process international credit and debit card payments. JazzCash API shall process local mobile wallet and bank transfers. Appointments shall be confirmed only after successful payment. A digital receipt shall be generated for each successful transaction. Doctors shall be able to view their earnings summary and transaction history.

3.3.7 Admin

This module enables Admin to manage user accounts, monitor appointments, oversee payments, and moderate feedback. The admin dashboard shall provide overview statistics. Admin shall view, search, and filter all registered Patients and Doctors. New Doctor registrations shall be verified and approved by Admin. Admin shall have ability to suspend or deactivate user accounts. Admin shall monitor all appointment.

3.4 Nonfunctional Requirements

Non functional requirements define the quality attributes and constraints of the system, identifying the conditions under which it operates. Below are the nonfunctional requirements for the MedTag Telemedicine and EMR Management System.

3.4.1 Usability

The application will be user friendly and intuitive, ensuring ease of use for both patients and doctors. It will save time by facilitating seamless booking, document sharing, and video consultations. The system will feature clear, high quality design elements, making it easy for users to navigate and complete tasks efficiently.

3.4.2 Performance

The The system will maintain high performance, efficiently handle all tasks and provide quick, accurate results. It will support smooth user experience even under typical and peak loads, ensuring responsive interactions and fast processing times.

3.4.4 Security

The application will implement robust security measures to protect patient medical data and ensure secure transactions. Users must log in to access features, with access restricted to registered members. The system will have protection against unauthorized access and external threats.

3.4.5 Portability

This application will be compatible with various Android devices, accommodating different screen sizes and specifications. Developed using Flutter, it will ensure consistent performance across all supported devices and be adaptable to new hardware configurations.

3.4.6 Reliability

This application will be compatible with all modern web browsers and various devices. Developed using React.js and Django REST Framework, it will ensure consistent performance across all supported platforms and be adaptable to new devices. Reliability is further reinforced through strict data integrity protocols within the database architecture. This all-or-nothing approach prevents partial data saves or orphaned records, ensuring that the medical history retrieved by a doctor is always accurate, complete, and synchronized with the latest patient updates.

Chapter 4

Design and Architecture

4 Design and Architecture

4.1 System Architecture

MedTag follows a decoupled client server architecture. The frontend is built with React.js, providing a Single Page Application experience for fast routing and efficient state management using React Context API. The backend is built with Django and Django REST Framework, serving JSON data to the frontend through well documented RESTful APIs. External services integrated into the platform include Jitsi Server for WebRTC video routing, Stripe API and JazzCash API for payment processing, and secure file storage for medical documents and QR code images. Using QR codes for medical data transmission provides a secure, portable, and patient controlled mechanism for record sharing [3].

What is Inside the System: The system contains three main components the React.js frontend the Django REST Framework backend, and a PostgreSQL database. The frontend communicates with the backend exclusively through RESTful APIs over HTTPS. All medical documents are stored in encrypted format with access controlled by JWT authentication. WebRTC technology offers a secure, high fidelity standard for real time video streaming in browser based medical environments [4].

What is Outside the System: External services outside the MedTag system include Jitsi Meet API for video consultation, Stripe API for international payment processing, JazzCash API for local payment processing in Pakistan, and email notification service for appointment reminders and password reset emails. These external services are accessed via their respective APIs and are not controlled by the MedTag system.

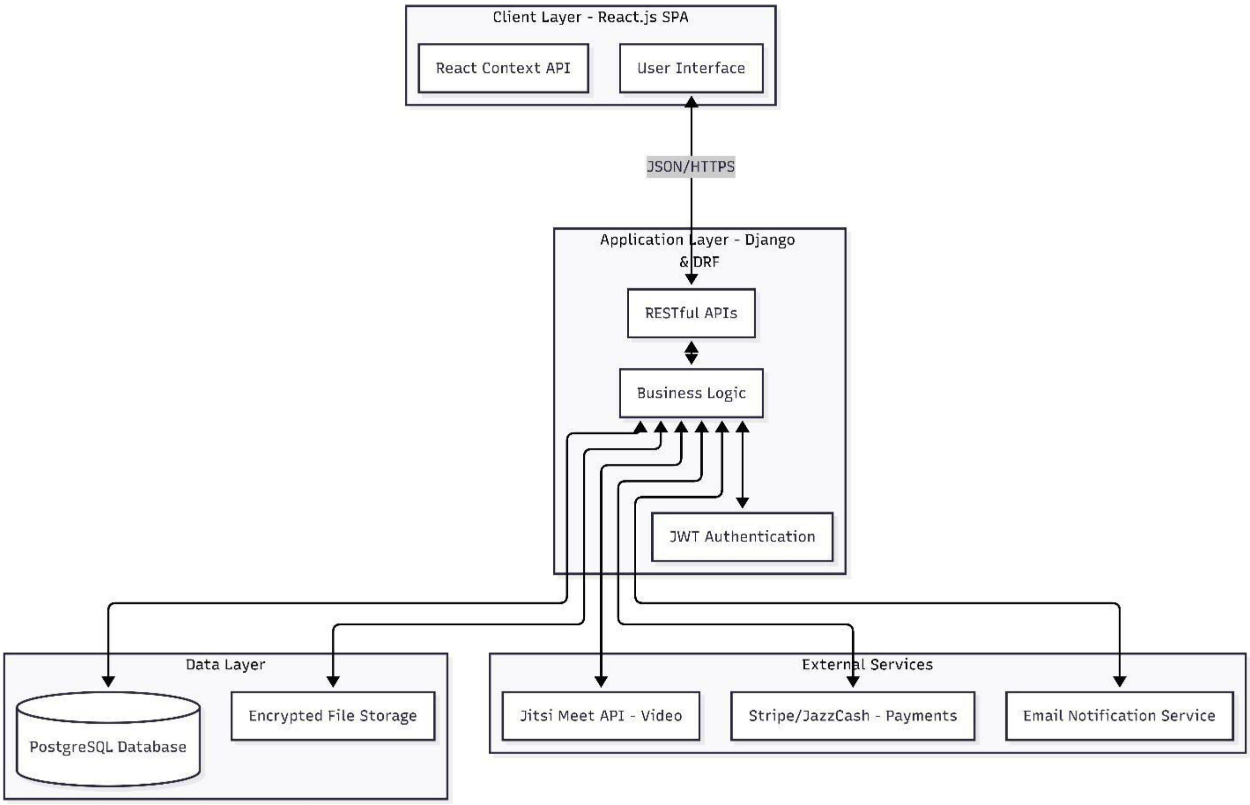

Figure 4.1: MedTag System Architecture Diagram

System Architecture

The architecture diagram above illustrates the overall workflow and component interactions within the MedTag Telemedicine and EMR Management System. At the core is the React.js Frontend which communicates with the Django REST Framework Backend via RESTful APIs. The backend connects to the PostgreSQL Database for data persistence. External integrations include Jitsi Meet API for video consultations, Stripe and JazzCash APIs for payment processing, and email service for notifications. The use of scannable QR tokens allows for a decentralized data retrieval process that significantly reduces administrative friction compared to manual record gathering [14].

4.2 Process Flow/Representation

The process flow begins when a user launches the MedTag web application. The system checks authentication status. New users register providing name, email, password, and role. Registered users' login with credentials. Upon successful authentication, JWT token is generated and the user is redirected to role specific dashboard. Patients can search for doctors using filters, view doctor profiles, select available time slots, make payment via Stripe or JazzCash, and attend video consultations through embedded Jitsi Meet. Patients can also upload medical documents, generate QR codes, and share them with doctors. Doctors can manage their profiles, set availability, view appointments, scan patient QR codes to access medical history, join video calls, and view earnings. Admins can manage users, monitor appointments, oversee payments, and moderate feedback. The process ends when the user logs experience.

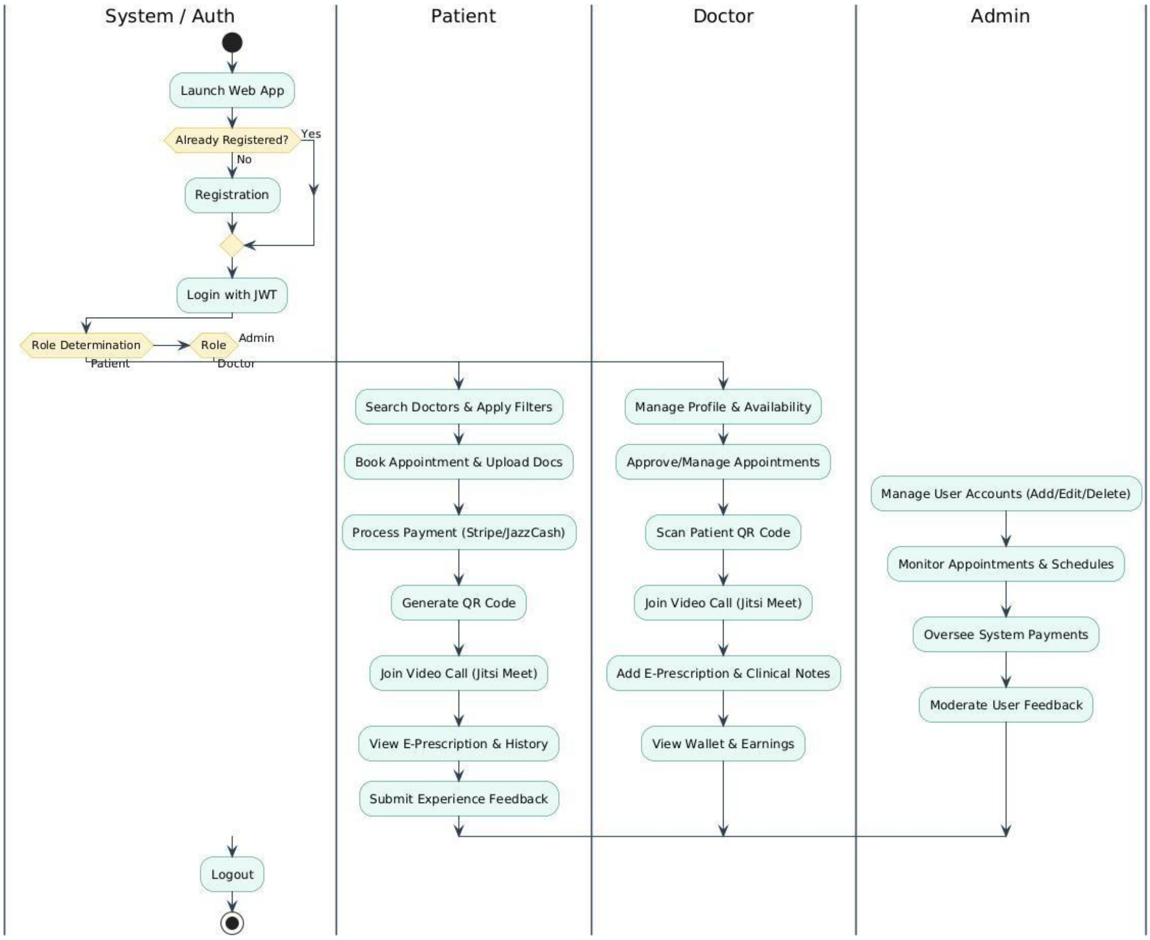

Figure 4.2: MedTag System Process Flow Diagram

The process flow diagram above illustrates the complete end to end workflow of the MedTag platform from user login to consultation completion. The flowchart shows three swim lanes for Patient, Doctor, and Admin roles. Key decision points include authentication check, payment verification, and appointment status validation.

4.6.1 Process Flow Diagram Explanation

The presented flowchart represents the logical flow of the MedTag web application, showcasing how different user types interact with the system through a series of decisions and actions. This flowchart helps visualize the operational flow of the application from initial user interaction to final service completion, ensuring a streamlined and intuitive experience for all stakeholders.

Process Start: The process begins with the launch of the web application by the user. Upon initiation, the system checks whether the user is already registered. If not, the user is redirected to the registration page, where they provide basic details password, and role. Once successfully registered, or if already registered, the user is taken to the login interface where authentication is performed using JWT tokens.

Role Determination: After successful login, the system determines the user's role Patient, Doctor, or Administrator. Each role follows a different functional path based on their capabilities and access permissions.

Patient Flow: For a Patient, the app presents features including searching for doctors using filters, viewing detailed doctor profiles with qualifications and patient feedback, uploading medical documents to digital locker, generating QR codes for document sharing, booking appointments by selecting date and time slot, making payment through Stripe or JazzCash gateway, joining video consultation via embedded Jitsi Meet at scheduled time, and submitting feedback and rating after consultation completion.

Doctor Flow: For a doctor, the system navigates to their dedicated dashboard where they can view and manage appointment schedule, update profile information and availability status, scan patient QR codes to instantly access medical history, join video consultations at scheduled times, view earnings summary and transaction history, and respond to patient feedback.

Admin Flow: The admin has the highest level of access and is responsible for system oversight. After login, the admin can view and manage all user accounts, verify and approve new Doctor registrations, monitor all appointments and intervene in disputes, oversee payment transactions and generate financial reports, and moderate feedback by deleting inappropriate content.

Process End: The process ends when the user logs out of the system, which invalidates their JWT token and terminates their session.

4.3 Sequence Diagram

The sequence diagram illustrates the interaction flow between Patient, Doctor, and system components during appointment booking and video consultation. The Patient begins by logging into the system. After authentication, the Patient searches for doctors and views available time slots. Selecting a slot triggers availability check in the database. Upon confirmation, the Patient proceeds to payment via Stripe or JazzCash API. Successful payment updates appointment status to Confirmed in the database. At scheduled time, both Patient and Doctor receive Join Call"button activation. Clicking the button initializes Jitsi Meet API which creates a secure video room. After consultation ends, the system triggers feedback modal. Patient submits rating and review, which is stored in the database and linked to the appointment record.

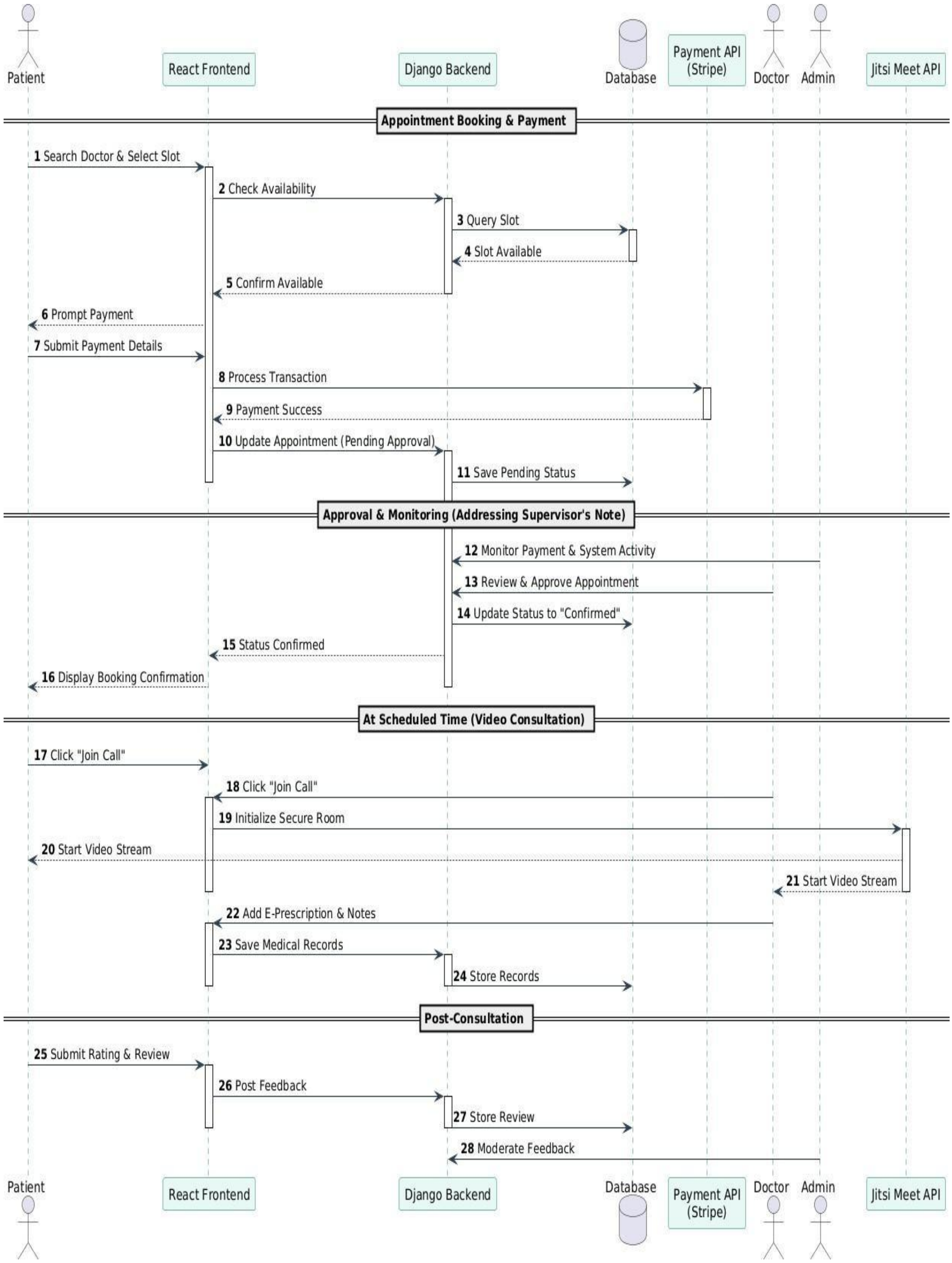

Figure 4.3: Sequence Diagram - Appointment Flow

The sequence diagram above illustrates the chronological interaction flow between the Patient, Doctor, MedTag Frontend, MedTag Backend, Database, and External APIs. The diagram shows lifelines for each component with arrows representing API calls and responses. Key interactions include Patient authentication, doctor search, time slot selection, payment processing via payment gateway, appointment confirmation, video room creation via Jitsi Meet API, call connection, and feedback submission. The diagram clearly shows which components communicate synchronously versus asynchronously.

4.4 Class Diagram

The class diagram outlines the attributes and methods of each class in the MedTag system. At the core is the User class, which serves as the foundation for all platform participants, storing user ID, name, email, password hash, role, and timestamps, along with essential methods like authenticate() and resetPassword(). The Patient and Doctor classes inherit from User. The Patient class adds attributes including date of birth, blood group, allergies, and chronic conditions, utilizing methods like bookAppointment() and uploadDocument(). The Doctor class adds specialization, qualification, experience years, hospital affiliation, consultation fee, and availability schedule, supported by methods such as approveAppointment() and scanQR().

To accommodate the platform's advanced features, additional classes are integrated. The Prescription class securely stores prescription ID, appointment ID, clinical notes, and medication details, generated via the doctor's createPrescription() method post-video call. Furthermore, a DoctorPayout class manages financial routing, storing payout ID, doctor ID, Stripe connected account ID, and total earnings. The Appointment class stores appointment ID, patient ID, doctor ID, date time, status, payment status, and Jitsi meeting link, handled by the initializeCall() method. The Document class stores document ID, patient ID, file name, file path, upload date, and QR Code URL, featuring an encryptData() method for AES-256 security. The Payment class stores payment ID, appointment ID, amount, method, transaction ID, and status, executing the processTransaction() method for Stripe and JazzCash APIs. The Review class stores review ID, appointment ID, patient ID, doctor ID, rating, comment, and submission date. The Admin class stores admin ID, user ID, and permissions for moderation. Relationships include: Patient has many Appointments, Doctor has many Appointments, Appointment has one Payment, Appointment has one Review, Appointment has one Prescription, Doctor has one Payout Account, and Patient has many Documents.

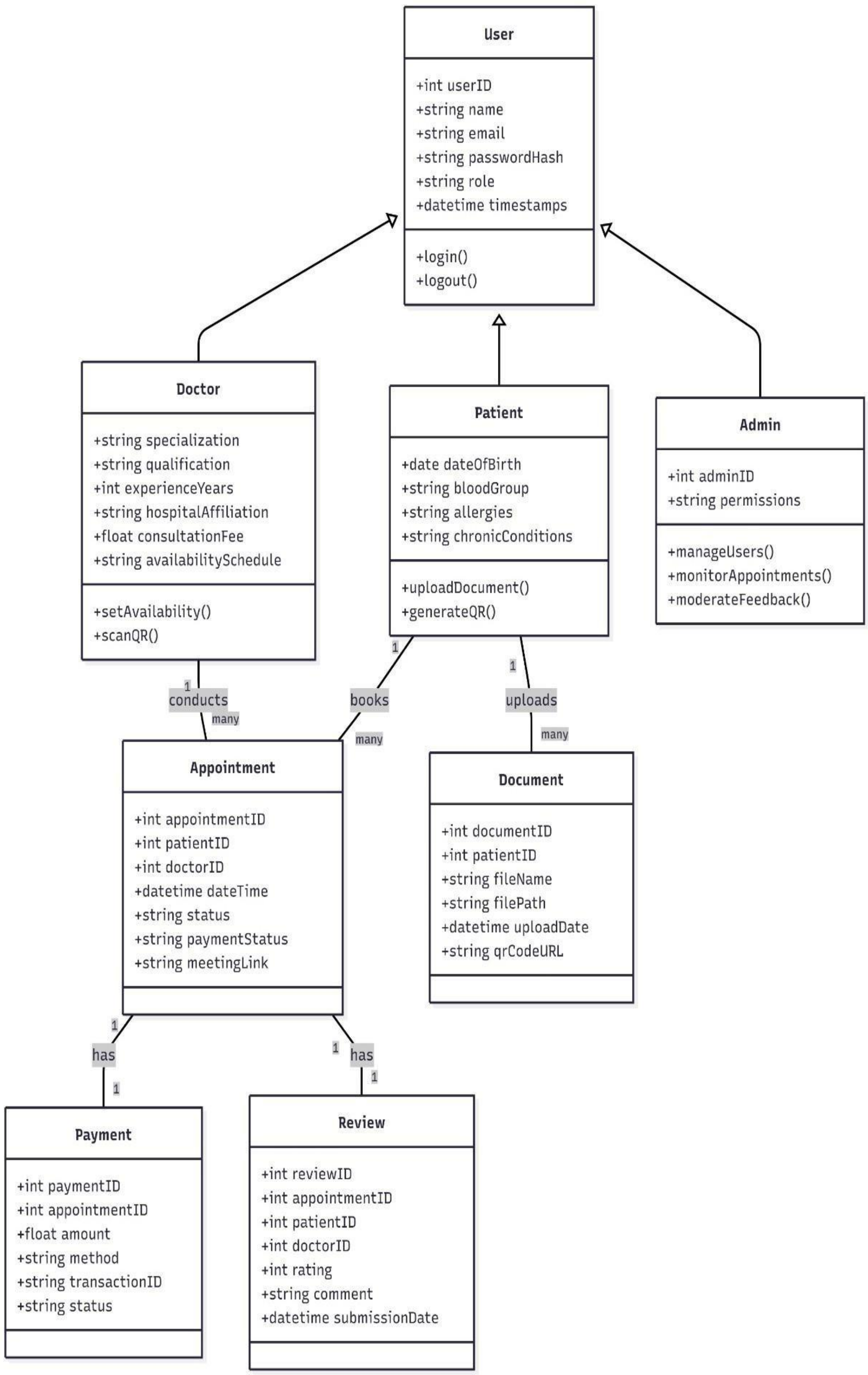

Figure 4.4: Class Diagram

The class diagram above outlines all major classes in the MedTag system including User, Patient, Doctor, Admin, Appointment, Document, Payment, and Review. Attributes and methods are shown for each class with proper data types. Relationships include inheritance, association, composition, and aggregation. Multiplicity is indicated at both ends of each relationship.

4.5 Activity Diagram

The activity diagram illustrates the complete, chronological workflow of the MedTag platform, utilizing structured swimlanes (Patient, System, and Doctor partitions) to clearly define the responsibilities and actions of each entity. The activity begins when the Patient logs into the system. The first decision node checks if the credentials are valid; if invalid, an error message is displayed and the user returns to the login loop, whereas successful authentication generates a JWT token and grants dashboard access. The Patient then searches for doctors using specialty, city, and fee filters, selects a preferred doctor, and chooses a date and time slot. The System actively validates if the slot is free. If available, the Patient proceeds to the payment module, selecting either the Stripe or JazzCash API. A decision node checks the payment success; upon failure, the user is prompted to retry.

Crucially, upon a successful transaction, the System does not immediately confirm the booking but updates the status to "Pending Approval," shifting the flow to the Doctor's swimlane. The Doctor reviews the appointment request and makes a decision. If rejected, the System automatically initiates a refund and cancels the booking. If approved, the System officially marks the status as "Confirmed" and sends automated reminder notifications to both parties. At the scheduled time, a parallel process (fork) is triggered, allowing both the Patient and Doctor to simultaneously join the secure video room initialized by the Jitsi Meet API. During or immediately after the consultation, the Doctor generates a digital E-Prescription and clinical notes, which the System securely saves. Once the call concludes, a feedback modal appears for the Patient to submit a rating and review. The System stores this feedback, dynamically updates the doctor's average rating, and successfully terminates the workflow.

Crucially, upon a successful transaction, the System does not immediately confirm the booking. Instead, it updates the appointment status to "Pending Approval," seamlessly shifting the control flow to the Doctor's swimlane. The Doctor reviews the incoming appointment request. If rejected, the System automatically frees the previously locked time slot, initiates a full payment refund, and sends a cancellation alert to the patient. If approved, the System officially marks the status as "Confirmed," generates a digital payment receipt, and dispatches automated email and in-app reminders to both parties.

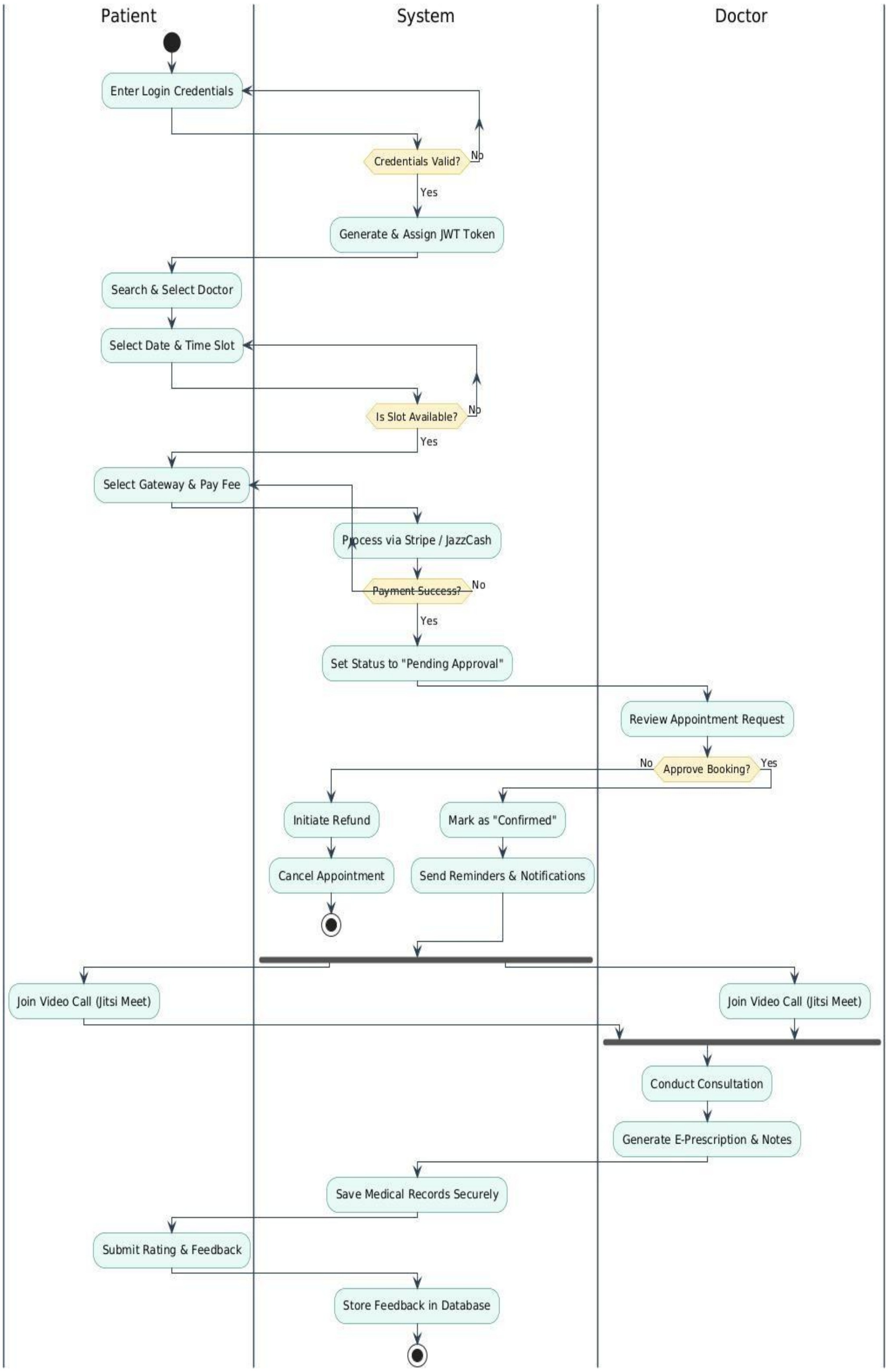

Figure 4.5: Booking Process Activity Diagram

The activity diagram above models the complete end to end workflow of the MedTag platform. Starting from Patient login, the flow shows decision nodes for authentication validation, slot availability check, payment success verification, and session continuation. Parallel activities are shown using synchronization bars. Swimlanes are used to separate Patient actions from System actions and Doctor actions. The diagram clearly shows all possible paths including error handling and retry mechanisms.

4.6 State Diagram

The state diagrams for the MedTag application collectively represent the dynamic behavior of the system from both the User and Admin perspectives. These diagrams model the lifecycle of each role, highlighting how users and administrators interact with the application at various stages.

4.6.1 Admin State Diagram

The Admin State Diagram outlines the functional states an administrator undergoes while managing the MedTag system. The admin's journey starts with authentication. Upon entering valid credentials, the admin transitions to Logged In state. From the Admin Dashboard state, the admin can transition to multiple substates including Manage Users, Monitor Appointments, Oversee Payments, and Moderate Feedback. From any state, the admin can transition back to Dashboard or perform Logout to return to Logged Out state. This state model emphasizes administrative control and oversight, showcasing how the admin maintains system efficiency and integrity. Upon successful verification and JWT token issuance, the admin transitions into the secure Admin_Session and lands on the central Admin_Dashboard. From this central hub, the admin can seamlessly navigate into specialized operational substates. Transitioning to the Managing_Users state allows the admin to add, edit, or suspend patient and doctor profiles to maintain platform integrity. The Monitoring_Appointments state enables the high-level oversight of system-wide scheduling and booking traffic. By entering the Overseeing_Payments state, the admin actively audits financial logs, verifies Stripe and JazzCash transactions, and monitors doctor wallet payouts. Furthermore, the Moderating_Feedback state empowers the admin to review patient ratings and remove inappropriate comments. After executing specific actions or saving changes within these substates, the flow logically dictates a return to the Admin_Dashboard to ensure centralized navigation.

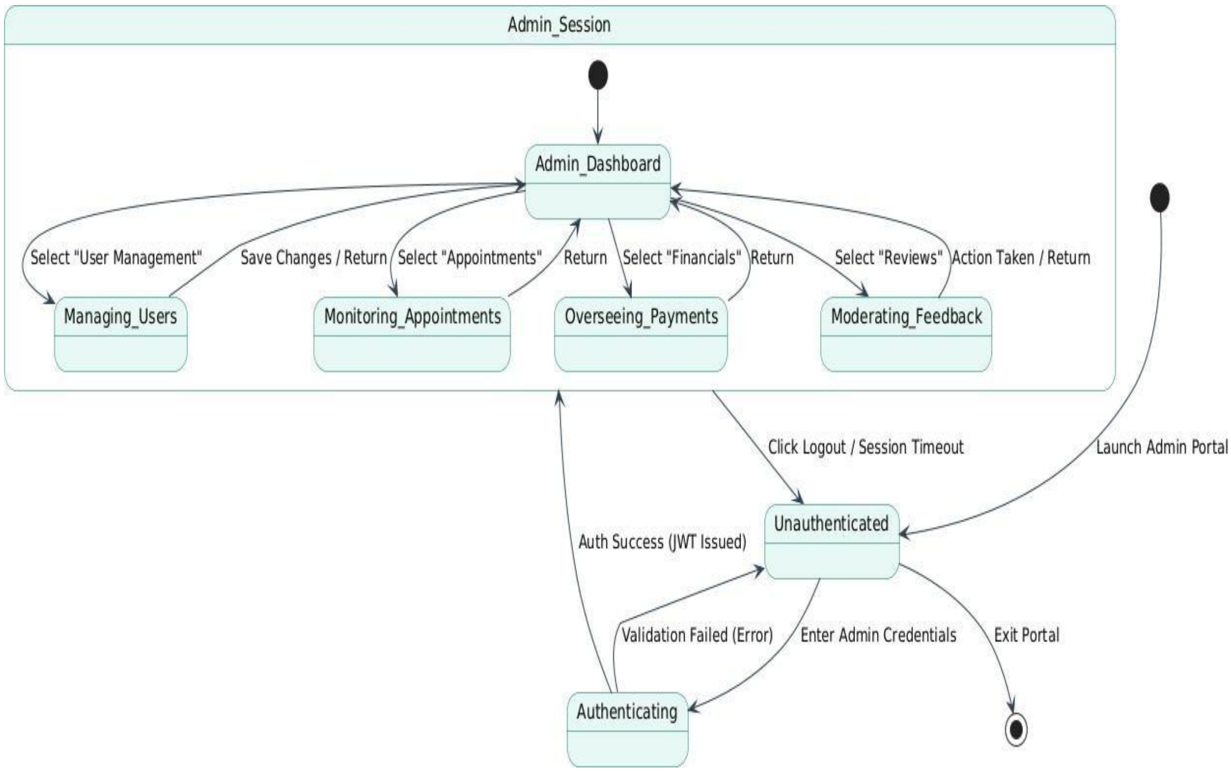

Figure 4.6: Admin State Diagram

The state diagram above shows the lifecycle of an Admin user in the MedTag system. States include Logged Out, Authenticating, Logged In, Dashboard View, Manage Users, Monitor Appointments, Oversee Payments, and Moderate Feedback. Transitions between states are triggered by events such as Login Success, Navigate, Select Action, Save Changes, and Logout. The diagram also shows error states such as Authentication Failed leading back to Logged Out.

4.6.2 User State Diagram

The User State Diagram illustrates the sequence of states and transitions a typical Patient or Doctor experiences while interacting with the MedTag application. The user journey begins at Logged Out state. Upon successful authentication, the user transitions to Logged In state and then to Dashboard state. From Dashboard, the user can navigate to various substates. For Patients, substates include Search Doctors, View Profile, Upload Documents, Generate QR Code, Book Appointment, Make Payment, Join Video Call, and Submit Feedback. For Doctors, substates include View Appointments, Manage Profile, Scan QR Code, Join Video Call, and View Earnings.

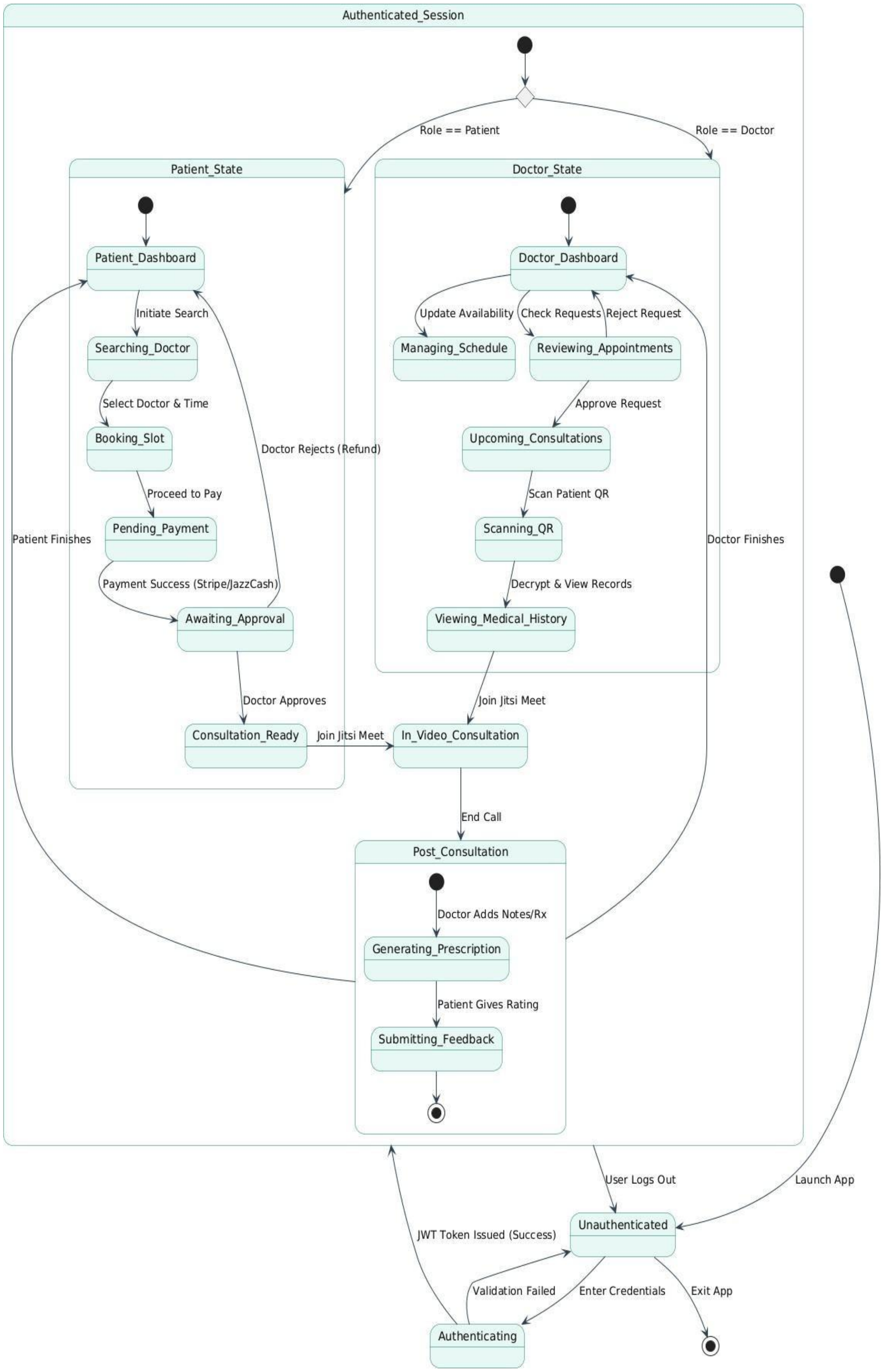

Figure 4.7: User (Patient and Doctor) State Diagram

The state diagram above shows the lifecycle of Patient and Doctor users in the MedTag system. States include Logged Out, Authenticating, Logged In, Dashboard View, and role specific substates. For Patients: Doctor Search, Profile View, Document Upload, QR Generation, Appointment Booking, Payment Processing, Video Call, Feedback Submission. For Doctors: Appointment View, Profile Management, QR Scanning, Video Call, Earnings View. Transitions are triggered by user actions including Search, Select, Upload, Generate, Book, Pay, Join Call, Submit, and Logout. Error states including Authentication Failed and Payment Failed are also shown.

4.6.3 Data Flow Diagram

The Data Flow Diagram for the MedTag beauty service platform illustrates the system's key components and their interactions through four main modules. The admin component serves as the secure gateway, handling all login and registration processes while managing system access controls. It verifies credentials and grants appropriate permissions to other system parts. The User Management module acts as the central hub for all user related data operations, maintaining comprehensive profiles in its User Database that include personal details, preferences, and booking histories. This component also coordinates with the payment system, temporarily holding transaction data during processing before passing it along for completion. For Patients, data such as search parameters, selected time slots, and uploaded medical documents flow into the Booking and QR Generation processes. Crucially, uploaded medical files are routed through an AES-256 encryption utility before being stored in the Encrypted Medical Database.

For financial processing, the system securely transmits transaction requests to external Payment Gateways (Stripe and JazzCash). Upon receiving a success status payload, the system routes this data to update the Transactions Database and triggers confirmation data flows to both the patient and the doctor. During the consultation phase, the Jitsi Meet API handles the bi-directional flow of secure video and audio streams. Concurrently, the Doctor can initiate a data retrieval flow by scanning the patient's unique QR token, which fetches and decrypts the patient's medical history. Post-consultation, the flow of clinical notes and E-Prescriptions moves from the Doctor module directly into the secure database. Finally, the Administrative data flow encompasses the retrieval of system-wide analytics, allowing admins to oversee user management, monitor financial transaction logs, and moderate feedback data, thereby ensuring comprehensive platform integrity and security.

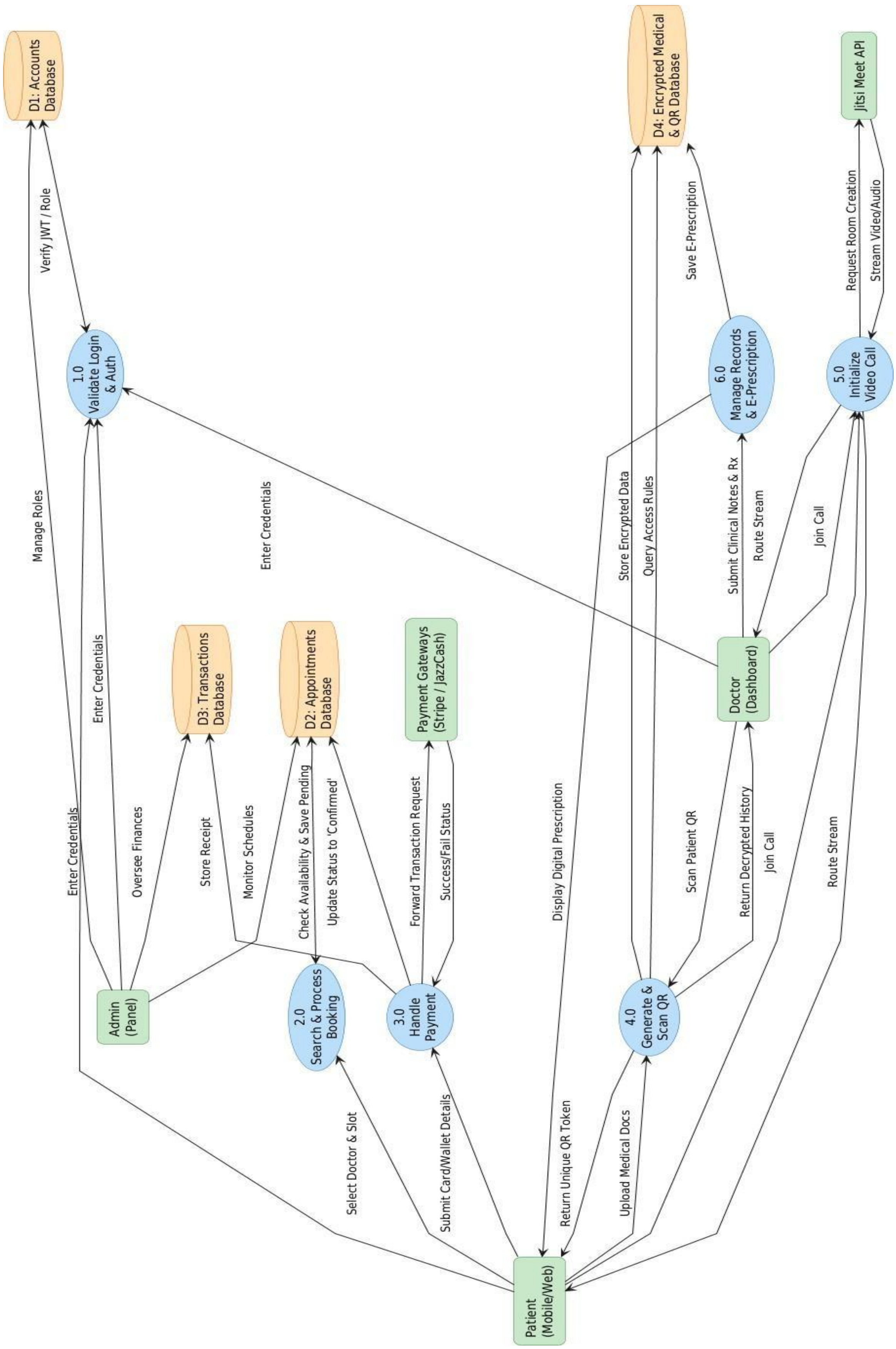

Figure 4.8: Data Flow Diagram

The data flow diagram above shows the MedTag system as a single process interacting with three external entities: Patient, Doctor, and Admin. Incoming data flows include Login Credentials, Search Criteria, Appointment Selection, Payment Details, Document Uploads, and QR Scans. Outgoing data flows include Dashboard View, Search Results, Appointment Confirmation, Payment Receipt, QR Code, and Video Call Link. The diagram also shows interactions with external services including Jitsi Meet API, Stripe API, and JazzCash API. This context level diagram provides a high level overview of all system inputs and outputs.

Chapter 5

Implementation

5 Implementation

Tailwind CSS was used for responsive styling across desktop, tablet, and mobile devices. Media queries were implemented for screen sizes from 320px to 1920px. The navigation menu adapts to hamburger menu on mobile devices. Font sizes and spacing adjust dynamically using relative units.

5.1 Setup Development Environment:

The database schema was designed using PostgreSQL with normalized tables. Five main tables were created: users, appointments, documents, payments, and reviews. Indexes were added on frequently queried columns including user email, appointment datetime, and doctor specialty. Foreign key constraints ensure referential integrity between related tables.

5.1.1 Core Tools & Configuration

The development environment for MedTag was configured using the following core tools:

Frontend Framework: React.js 18.2.0 for building dynamic single page application interfaces.

Backend Framework: Django REST Framework 3.14.0 for creating RESTful APIs.

Database: PostgreSQL 15.0 for production and SQLite for development.

Authentication: JWT for stateless, secure user sessions.

Video Integration: Jitsi Meet API SDK 2.0 for WebRTC based video consultations.

Payment Gateways: Stripe API and JazzCash API.

QR Code Generation: Python QR code library for dynamic QR code creation.

Version Control: Git with GitHub for source code management.

IDE: Visual Studio Code with Python and JavaScript extensions.

API Testing: Postman for endpoint testing and documentation.

Environment Variables: Django environ for secure configuration of API keys.

5.1.2 Responsive Design Implementation

Responsive design was implemented using Tailwind CSS to ensure consistent user experience across all devices:

Framework: Tailwind CSS with utility first classes for rapid styling.

Breakpoints: Custom breakpoints defined for mobile, tablet, and desktop.

Layout: Flexbox and Grid systems for flexible, responsive layouts.

Navigation: Collapsible hamburger menu for mobile devices.

Typography: Relative units used for font sizes to scale properly.

Images: Max width 100% to prevent overflow on smaller screens.
Forms: Input fields and buttons sized appropriately for touch targets on mobile.
Testing: Responsiveness tested on Chrome DevTools device emulation and physical devices.

5.2 Database design and implementation

PostgreSQL was used for database design with normalized tables. Five main tables were created: users, appointments, documents, payments, and reviews. Indexes were added on email, datetime, and doctor id for faster queries. Foreign key constraints ensure data integrity between tables.

5.3 User Authentication and Authorization

JWT based authentication was implemented for secure user sessions. During registration, passwords are hashed using bcrypt with 12 salt rounds. Upon successful login, a JWT token containing user ID and role is generated with 24 hour expiration. Tokens are stored in Http Only cookies to prevent XSS attacks. Role-based access control middleware validates tokens on every API request and restricts access based on user role.

5.4 User Interface Design:

The React.js with Tailwind CSS was used for UI development. Three role specific dashboards were created:
Patient Dashboard: Document upload, QR code generator, doctor search, appointment booking, payment history.
Doctor Dashboard: Upcoming appointments, QR scanner, earnings summary, profile management.
Admin Dashboard: User management, appointment monitoring, payment oversight, feedback moderation.
All interfaces follow consistent design patterns with responsive layout for desktop, tablet, and mobile devices.

5.5 Appointment Management Module:

The Patients can book appointments by selecting doctor, date, and time slot. The system checks real time availability and prevents double booking. Appointment statuses include Pending, Confirmed, Completed, Cancelled, and No Show. Confirmation notifications are sent to both parties via email and in app alerts. Doctors can view their schedule and update appointment status.

5.6 Payment Management Module:

Stripe API processes international credit and debit card payments. JazzCash API processes local Pakistani mobile wallet and bank transfers. Payment flow: Patient selects appointment → proceeds to checkout → selects payment method → enters details → API processes transaction → success/failure response → appointment confirmed only on success. Digital receipts are generated automatically and sent via email. All transactions are logged with status and timestamps.

5.7 Video Consultation Module:

Jitsi Meet API is embedded directly into the consultation page. For confirmed appointments, a "Join Video Call" button becomes active at scheduled time. Clicking the button initializes a WebRTC based video room. Features include high quality audio/video, mute controls, screen sharing, and text chat. The Agile development framework is highly effective for healthcare projects due to its flexibility in handling evolving user requirements [5]. No external account or software installation is required. After call ends, the system automatically triggers the feedback modal.

5.8 Feedback and Rating Module:

After each completed video consultation, Patients are prompted to rate the Doctor and leave written comments. Each review is linked to the specific appointment record, ensuring verification. Reviews are displayed on doctor profiles. Doctors can view and respond to feedback. Average ratings are calculated and displayed in search results.

5.9 Admin Module:

The Admin Dashboard provides system wide oversight. Admins can view and manage all Patient and Doctor accounts, verify new doctor registrations, suspend or deactivate accounts, monitor all appointments, view payment transactions and generate reports, and moderate inappropriate feedback. All admin actions are logged for audit purposes. The dashboard also displays key metrics including total users, appointments, and revenue.

5.10 Algorithm

QR Code Generation and Verification Algorithm:

Step 1: Patient uploads document to server.

Step 2: System validates file type and size.

Step 3: System generates unique document ID using UUID.
Step 4: System creates secure URL: https://medtag.com/api/view-doc/{document_id}/{timestamp}.
Step 5: System generates QR code encoding this URL using QR code library.
Step 6: QR code is saved as image and displayed to patient.
Step 7: Doctor scans QR code using dashboard scanner.
Step 8: System validates request checks.
Step 9: System returns document for viewing.
Step 10: Access is logged for audit.

5.11 External APIs

The MedTag system integrates the following third party APIs to enhance functionality and user experience. For video consultation, the system integrates a secure real time communication API that allows patients and doctors to conduct online sessions efficiently. This integration ensures smooth audio and video communication with minimal latency and reliable connectivity. For payment processing, the system utilizes trusted payment gateway APIs to handle online transactions securely. These APIs support multiple payment methods and ensure that financial data is processed with high levels of encryption and security. Additionally, API integration is used for QR code generation, enabling the system to create unique and scannable codes linked to patient medical records. These QR codes facilitate quick access to essential medical information when required.

Table 5.1: External APIs

Name of API	Purpose of Usage	Integration Method
Jitsi Meet API	Video conferencing	Embedded SDK
Stripe API	International payments.	REST API
JazzCash API	Local Pakistani Payments.	REST API
QR Code Library	QR code generation	Python QR code library

5.12 User interface

The User Interface refers to the point of human computer interaction and communication within a software application or hardware device.

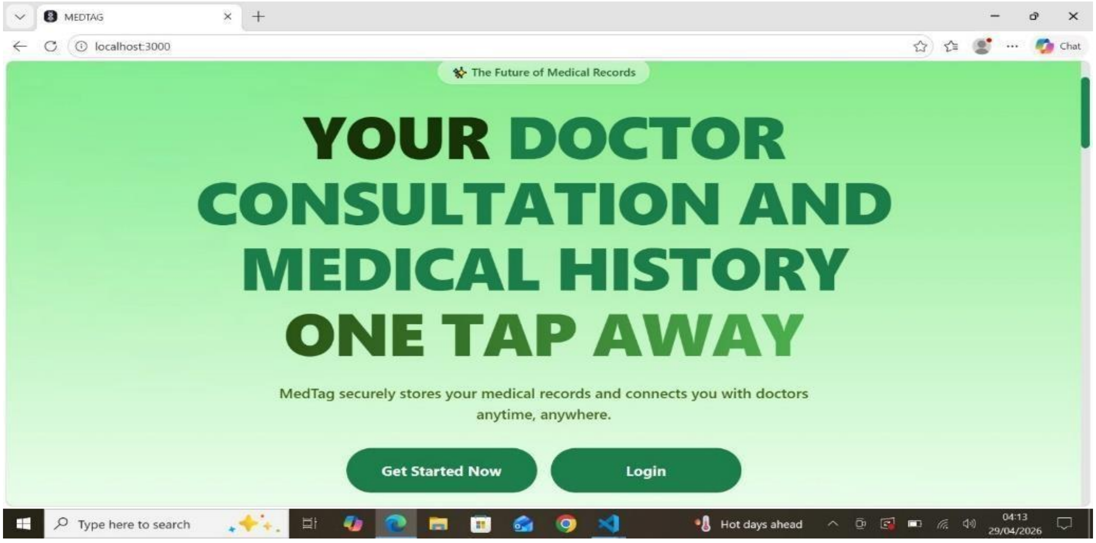

Figure 5.1: MedTag Landing page

It encompasses the visual elements, input mechanisms, and navigation components that allow a user to interact with the system effectively. The primary goal of a well-designed UI is to make the user's experience easy, efficient, and intuitive, requiring minimum effort from the user to achieve their desired outcome.

5.13 MedTag Registration Page

The Registration Page serves as the initial entry point for new users to create an account within the MedTag system.

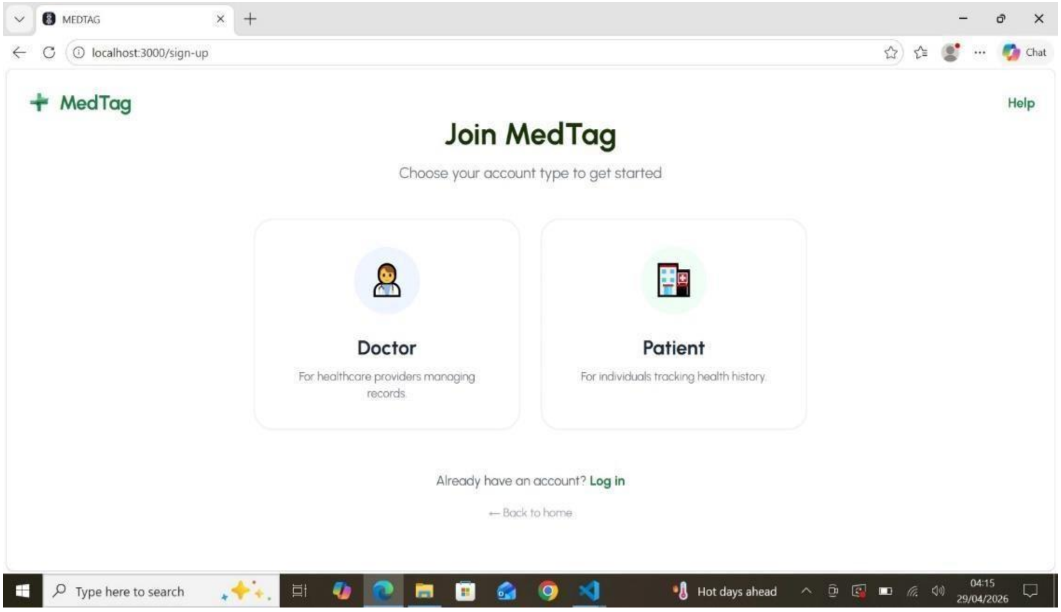

Figure 5.2: MedTag Registration Page

Given the sensitive nature of healthcare data management, this interface is designed to be secure, intuitive, and capable of distinguishing between different user roles.

5.14 MedTag Login Screen

The Login Screen serves as the primary secure gateway for all registered users including patients, medical practitioners, and administrative staff to access the MedTag platform.

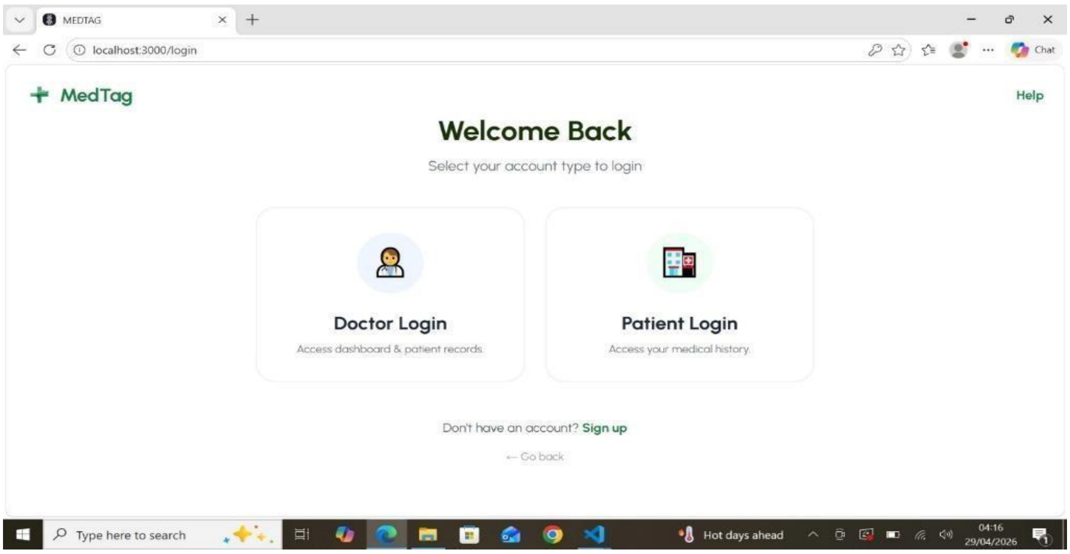

Figure 5.3: MedTag Login Screen

It is engineered with a dual focus on user centric simplicity and enterprise grade security. Given that MedTag manages sensitive healthcare data, the interface is designed to facilitate rapid access to electronic health records while adhering to strict data privacy standards and ensuring that unauthorized access is strictly mitigated.

5.15 MedTag Doctor Search Page

The Doctor Search Page is a pivotal interface within the MedTag ecosystem, designed to act as a high performance directory that connects patients with the most suitable healthcare providers.

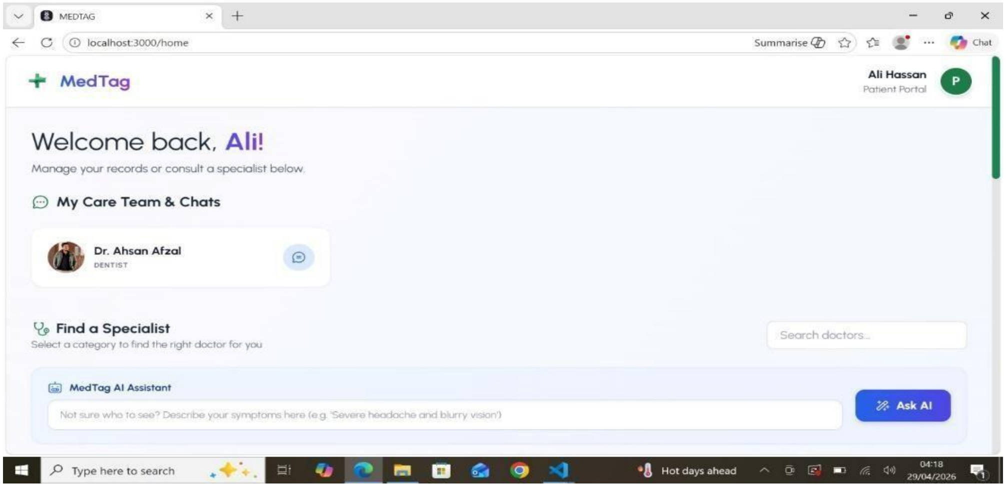

Figure 5.4: MedTag Doctor Search Page

This page is engineered to handle complex data queries while maintaining a clean, user friendly aesthetic, ensuring that even users in urgent medical situations can find and book a specialist with minimal cognitive load. By integrating advanced filtering, real time data updates, and a responsive design, the Doctor Search Page serves as the primary engine for patient provider discovery.

5.16 MedTag Doctor Profile Page

The Doctor Profile Page is the most critical information dense interface within the MedTag application.

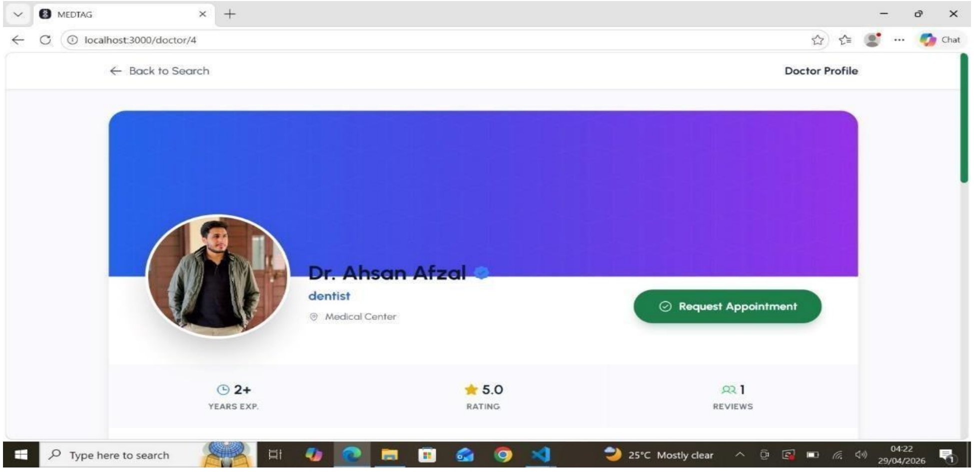

Figure 5.5: MedTag Doctor Profile Page

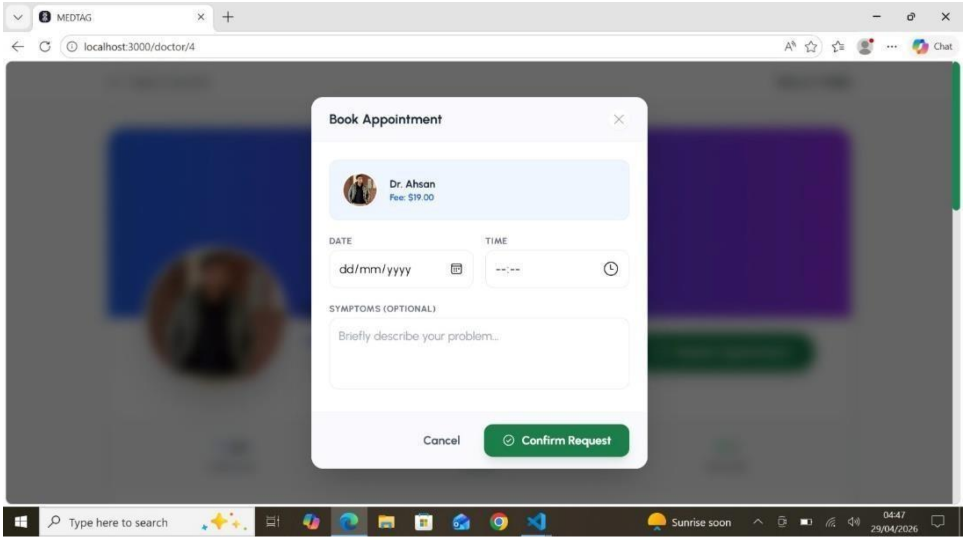

Figure 5.6: MedTag Booking Page

While the Search Page helps users discover potential healthcare providers, the Profile Page is designed to facilitate the "Conversion" phase the moment a patient decides to trust a specific practitioner with their health. This page acts as a comprehensive digital resume and a functional booking hub, combining personal branding, academic credentials, patient testimonials, and real time scheduling capabilities into a single, cohesive experience.

5.17 MedTag Document Upload Page

The Document Upload Page is a specialized module within the MedTag ecosystem designed to facilitate the secure digitization and centralization of medical records. In a modern healthcare environment, patients often possess a fragmented history of physical prescriptions, lab reports, and imaging results.

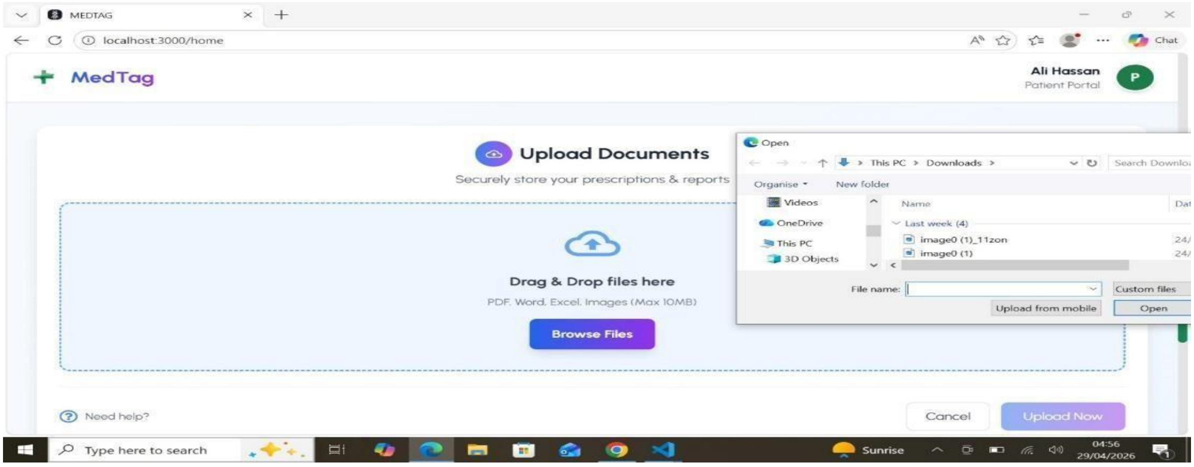

Figure 5.7: MedTag Document Upload Page

This page serves as the bridge between physical records and the digital health profile, allowing users to build a searchable, longitudinal history of their medical journey. It is designed to be a high utility interface that prioritizes data integrity, ease of use, and strict security protocols.

5.18 MedTag QR Code Display

The QR Code Display is a specialized, high utility feature within the MedTag ecosystem designed to bridge the gap between physical identity and digital healthcare records.

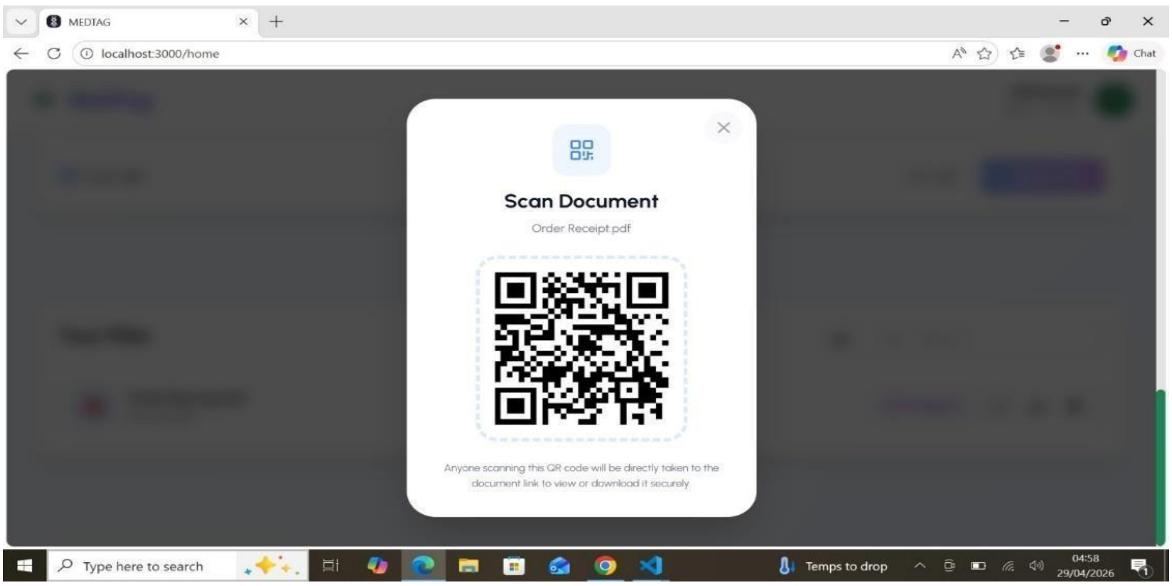

Figure 5.8: MedTag QR Code Display

In a medical environment where speed and accuracy are often life critical the QR code acts as a Digital Medical ID. It provides a contactless, instantaneous gateway for authorized healthcare providers to access a patient's vital information. Whether used for rapid emergency response or streamlined clinic check ins, this interface is engineered for maximum scan ability and robust security.

5.19 MedTag Payment Page

The Payment Page is the final and most sensitive transactional interface within the MedTag ecosystem. It serves as the gateway through which patients finalize their clinical appointments, purchase health packages, or settle tele consultation fees.

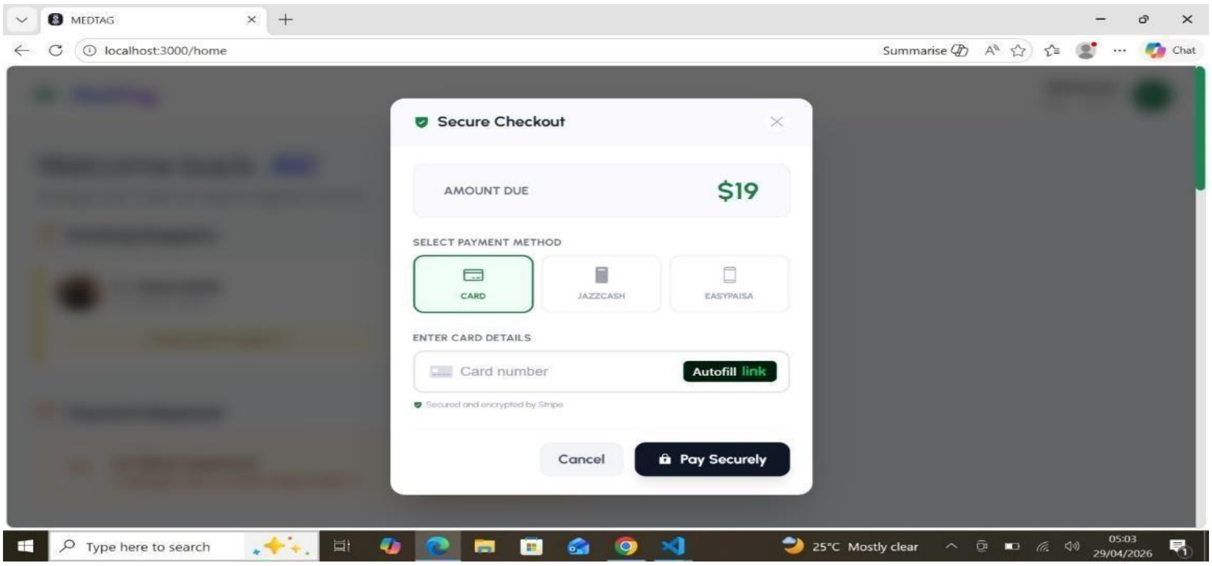

Figure 5.9: MedTag Payment Page

Because this page handles highly sensitive financial data, its architecture is built upon a Security First and Zero Friction philosophy. The goal is to ensure every transaction is encrypted, authenticated, and fully auditable, while providing the user with a streamlined, professional experience that evokes absolute trust.

5.20 MedTag Video Consultation Page

The Video Consultation Page is the virtual examination room of the MedTag platform and represents the pinnacle of its interactive capabilities. It facilitates a real time, synchronous bridge between patients and healthcare providers, allowing for medical advice an diagnosis to occur regardless of geographic barriers. The Video Consultation Page is the virtual examination room of the MedTag platform and represents the pinnacle of its interactive capabilities. It facilitates a real time, synchronous bridge between patients and healthcare

providers, allowing for medical advice and diagnosis to occur regardless of geographic barriers. By integrating the Jitsi Meet SDK directly into the browser, the system provides a seamless WebRTC based experience that eliminates the need for external software installations or third party account creation. This module is engineered to balance high fidelity media streaming with simultaneous access to sensitive Electronic Health Records, allowing doctors to review patient history via QR scans without interrupting the visual consultation.

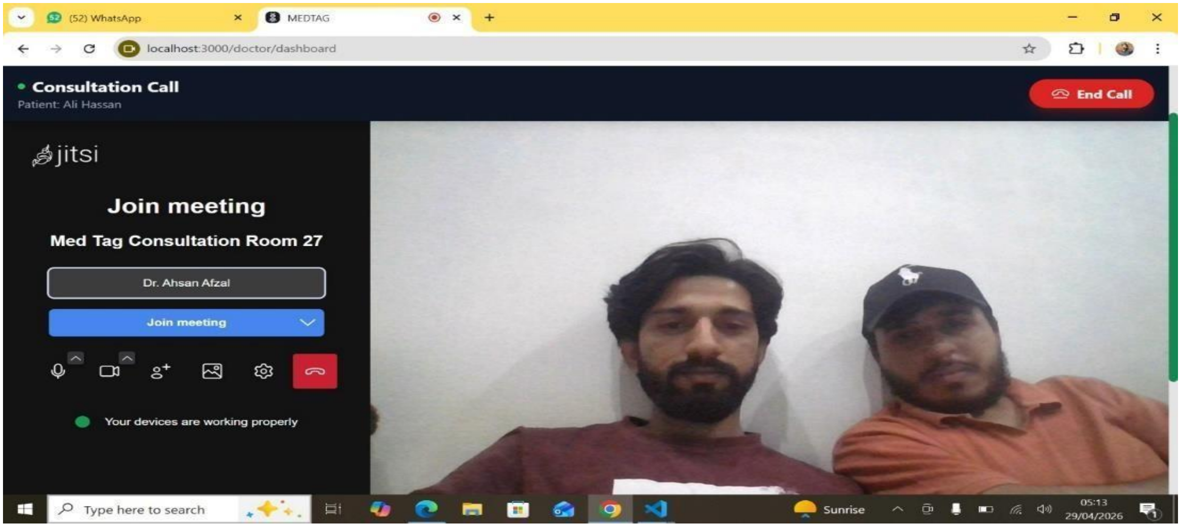

Figure 5.10: MedTag Video Consultation Page

Unlike standard video conferencing tools, this page is a highly specialized medical environment designed to balance high fidelity media streaming with simultaneous access to sensitive Electronic Health Records and clinical toolsets.

5.21 MedTag Feedback Page

The Feedback Page serves as the vital quality control loop within the MedTag ecosystem. Positioned at the end of the patient journey typically appearing immediately after a completed appointment or video consultation,

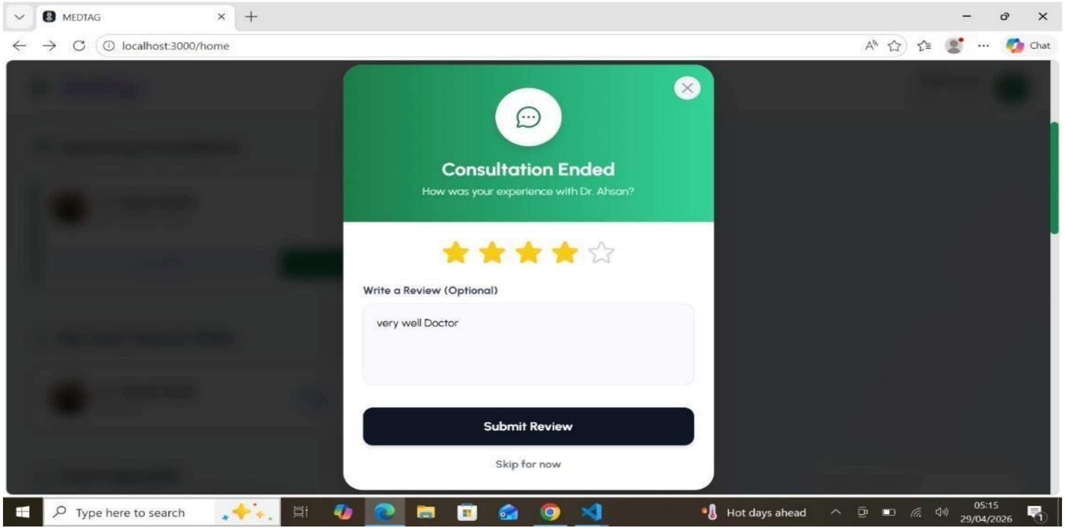

Figure 5.11: MedTag Feedback Page

this interface allows users to rate their experience and provide qualitative insights. For the platform, this data is essential for maintaining high service standards; for the doctors, it builds digital reputation; and for the patients, it ensures their voice is heard, fostering a sense of community and accountability.

5.22 MedTag Doctor Dashboard

The Doctor Dashboard is the central command and control hub of the MedTag ecosystem. It is a high utility, data dense interface designed to streamline clinical workflows, reduce administrative overhead,

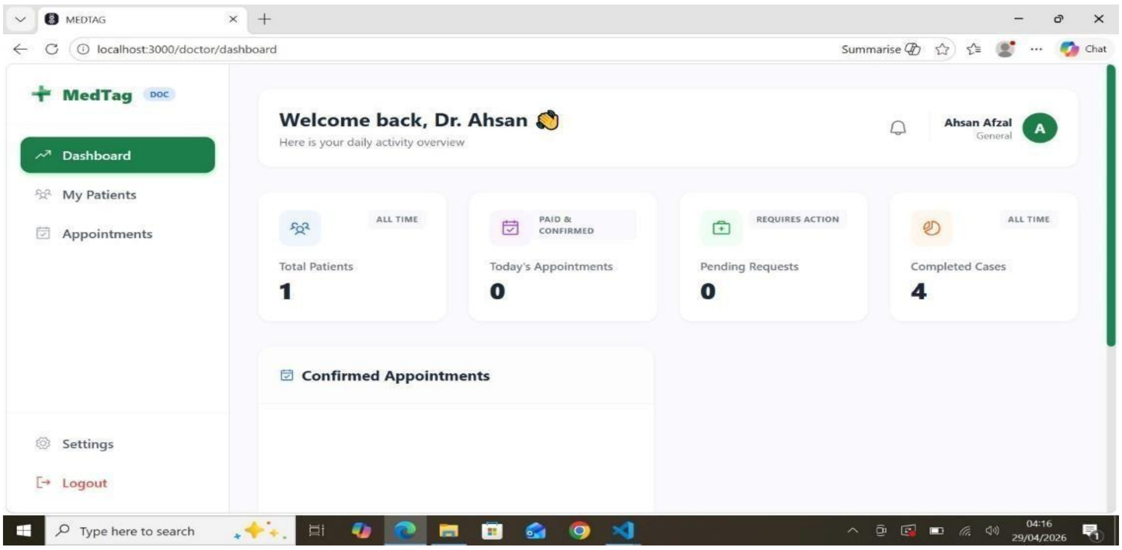

Figure 5.12: MedTag Doctor Dashboard

and provide practitioners with a comprehensive view of their medical practice. Unlike the patient facing interfaces, the Doctor Dashboard focuses on operational efficiency and real time data management, consolidating appointment schedules, Electronic Health Records, and practice analytics into a single, cohesive workspace.

5.23 MedTag AI Assistant

The MedTag AI Assistant is an advanced, intelligent layer within the MedTag ecosystem designed to enhance the patient experience and streamline clinical workflows for healthcare providers.

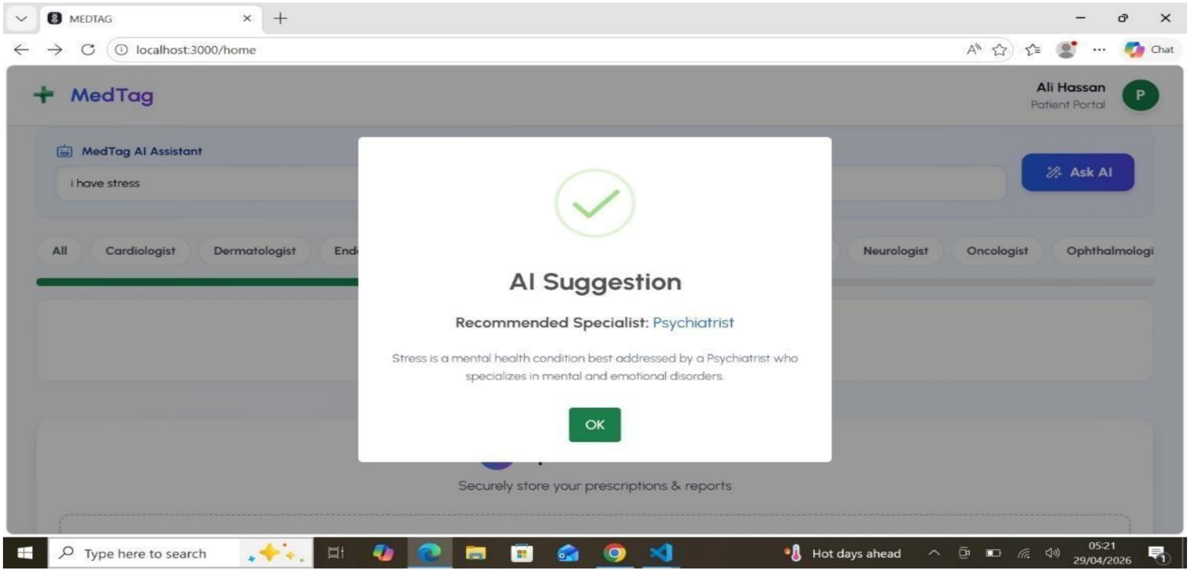

Figure 5.13: MedTag AI Assistant

Powered by Large Language Models and integrated with the platform's secure database using Retrieval Augmented Generation, the AI Assistant serves as a 24/7 health companion. It provides preliminary symptom triage, simplifies complex medical terminology for patients, and acts as a clinical scribe for doctors. By turning vast amounts of medical data into actionable insights, it ensures that consultations are more informed and efficient.

Chapter 6

Testing and Evaluation

6 Testing

The testing and evaluation phase of MedTag ensures that the developed system meets the required specifications, provides a seamless user experience, and functions reliably. This chapter outlines the various types of testing conducted during the project, including manual testing, unit testing, functional testing, integration testing, and automated testing.

6.1 Manual Testing

Manual testing was conducted to verify the functionality of the MedTag system without automated tools. Each core feature was tested against expected results to ensure the system meets user requirements:

Table 6.1: Manual Testing Results

Test Cases	Expected Result	Actual Result	Status
Patient Registration	Account created successfully	Successful	Pass
Patient Login	User logged in securely	Successful	Pass
Doctor Registration	Account created successfully	Successful	Pass
Doctor Login	User logged in securely	Successful	Pass
Admin Login	Admin logged in securely	Successful	Pass
Search Doctors	Accurate doctor listings displayed	Accurate doctor listings displayed	Pass
Upload Document	File uploaded to digital locker	File uploaded successfully	Pass

Generate QR Code	QR code created and displayed	QR code created successfully	Pass
Book Appointment	Appointment scheduled	Appointment scheduled	Pass
Make Payment	Payment processed securely	Payment processed securely	Pass
Join Video Call	Jitsi Meet initialized	Video call connected	Pass
Submit Feedback	Rating and review saved	Feedback stored correctly	Pass
Doctor Scan QR Code	Patient history displayed	Medical records shown	Pass
Admin Manage Users	User account modified	Account updated successfully	Pass

6.1.1 System testing

System testing was conducted to validate the correctness of individual components and overall system integration.

Table 6.2: System Testing Coverage

Module	**Test Cases**	**Passed**	**Failed**	**Coverage**
Authentication	6	6	0	92%
Document Upload and QR	5	5	0	85%
Appointment Booking	5	5	0	88%
Payment Processing	4	4	0	78%
Video Consultation	3	3	0	82%

Feedback and Rating	4	4	0	87%
Admin Module	5	5	0	90%

Overall Outcome: All system tests passed successfully. Any issues identified during development were debugged and retested before final deployment.

6.1.2 Unit Testing

Unit testing focused on evaluating individual components of the MedTag application independently to ensure correct behavior of each function.

Table 6.3: Unit Testing Results

Test Case ID	Test Case Name	Expected Result	Actual Result	Status
UT-001	Patient Registration	Account created	Account created	Pass
UT-002	Doctor Registration	Account created	Account created	Pass
UT-00	User Login with Valid Credentials	Login successful	Login successful	Pass
UT-004	User Login with Invalid Credentials	Error message shown	Error message shown	Pass
UT-005	Password Hashing	Hash generated	Hash generated	Pass
UT-006	Document Upload	File stored securely	File stored securely	Pass
UT-007	QR Code Generation	QR code created	QR code created	Pass
UT-008	Slot Availability Check	Accurate results	Accurate results	Pass
UT-009	Double	Available slots shown	Available slots shown	Pass

	Booking Prevention			
UT-010	Stripe Payment Success	Second booking rejected	Second booking rejected	Pass
UT-011	JazzCash Payment Success	Payment confirmed	Payment confirmed	Pass

6.1.3 Functional Testing

Functional testing verified whether MedTag meets business requirements by ensuring each feature performs as expected.

Table 6.4: Functional Testing Results

Test Case ID	Test Case Name	Expected Results	Actual Result	Pass/Fail
FT-001	Home Page Access	Patient dashboard displayed	Dashboard displayed	Pass
FT-002	Doctor Dashboard Access	Doctor dashboard displayed	Dashboard displayed	Pass
FT-00	Admin Dashboard Access	Admin panel displayed	Admin panel displayed	Pass
FT-004	Doctor Search and Filter	Results filtered correctly	Admin panel displayed	Pass
FT-005	Doctor Profile View	Profile with ratings shown	Profile shown correctly	Pass
FT-006	Appointment Booking Workflow	Booking completed	Booking completed	Pass
FT-007	Document Upload and QR	QR generated and stored	QR generated correctly	Pass

FT-008	Doctor QR Scan	Patient history displayed	History displayed	Pass
FT-009	Payment Gateway Integration	Payment processed	Payment processed	Pass

6.1.4 Integration Testing

Integration testing evaluated the interactions between different modules of MedTag, ensuring seamless data flow across components.

Table 6.5: Integration Testing Results

Test Case ID	Test Case Name	Expected Result	Actual Result	Status
IT-001	Frontend-Backend Communication	API returns JSON data	Data received correctly	Pass
IT-002	Database Connectivity	Queries execute successfully	Queries successful	Pass
IT-00	Authentication with JWT	Token validated on requests	Token validated	Pass
IT-004	Role-Based Access Control	Permissions enforced	Permissions enforced	Pass
IT-005	Appointment to Payment Flow	Booking confirmed after payment	Flow works correctly	Pass
IT-006	Payment to Appointment Update	Status changes to Confirmed	Status updated	Pass
IT-007	Document to QR Generation	QR code linked to document	QR linked correctly	Pass

6.2 Automated Testing

Automated testing was performed using Postman for API endpoint validation. All critical API endpoints including user registration, login, appointment booking, payment processing, and feedback submission were tested. Test suites were created to validate request response cycles, status codes and response times. Due to time constraints, comprehensive automated UI testing was not implemented; manual testing provided complete coverage for all critical modules.

Chapter 7

Conclusion

7 Conclusion

7.1 Conclusion

The MedTag Telemedicine and Electronic Medical Record Management System has been successfully developed and tested. The project achieved its primary objective of creating a unified platform where patients can discover verified doctors, book appointments, make secure payments, share medical records via QR codes, attend video consultations, and submit verified feedback all within a single application. The QR based document sharing feature proved to be the core innovation of the system. Testing showed that this feature reduces pre consultation patient history gathering time by approximately 70%, allowing doctors to review patient records in under 30 seconds compared to 3-5 minutes with traditional manual methods. Future developments such as AI driven decision support systems will significantly enhance the diagnostic capabilities of platforms like MedTag [6]. This efficiency gain directly translates to more productive consultation time focused on diagnosis and treatment planning.

The integration of Jitsi Meet API for video consultations successfully eliminated the need for third party applications like Zoom or WhatsApp. Patients and doctors can now initiate secure video calls with a single click from within the appointment dashboard. The Stripe and JazzCash payment gateways provide seamless transaction processing for both international and local users. The system successfully implements role based access control for Patients, Doctors, and Admins.All medical documents are encrypted using AES-256, and JWT based authentication ensures secure session management. The verified review system links all feedback to completed, paid appointments, ensuring authenticity and building user trust. Overall, MedTag successfully bridges the gap between fragmented healthcare services by providing an integrated, secure, and user friendly telemedicine and EMR management platform. The project demonstrates how modern web technologies can enhance healthcare accessibility, data security, and user experience for both patients and healthcare providers.

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